



2024 Regional Transmission Plan

December 2024

Executive Summary

The 2024 Regional Transmission Plan (RTP) is the result of a coordinated planning process performed by ERCOT Grid Planning with extensive review and input by Transmission Planners (TPs) and Transmission Owners (TOs) registered with North American Electric Reliability Corporation (NERC), and other stakeholders. The 2024 RTP addresses ERCOT System transmission needs for years 2026 through 2030. This report documents the results of the assessment, in part, to comply with the requirements of NERC Reliability Standards, ERCOT Protocols, and the ERCOT Planning Guide.

The reliability analysis was performed over a 6-year planning horizon; years 2 through 5 representing the near-term horizon, and year 6 representing the long-term horizon. The 2024 RTP assessed ERCOT's steady-state transmission needs under summer peak and off-peak conditions. In addition to the seasonal variations, the 2024 RTP also included sensitivity analyses to address uncertainty involved in the transmission planning process. The reliability analysis in the 2024 RTP included:

- Steady-state contingency analysis to identify criteria violations based on NERC Reliability Standards and ERCOT planning criteria;
- Short-circuit analysis to identify over-dutied circuit breakers in the near-term planning horizon; and
- Cascading analysis to identify potential system cascading conditions.

Following the reliability assessment, ERCOT, in collaboration with TPs, developed Corrective Action Plans (CAPs) to address the reliability criteria violations identified in this assessment. These plans included, but were not limited to, upgrades or additions of new transmission facilities and new Constraint Management Plans (CMPs).

The ERCOT System is experiencing rapid changes, including trends of notable growth in demand and penetration of intermittent Generation Resources. On August 10, 2023, ERCOT set the current all-time peak demand record of 85,508 MW, surpassing the previous record of 80,148 MW set in 2022. This trend of rising demand is expected to continue, driven by factors such as the further electrification of oil and gas processes in the Permian Basin and continued interest in connecting large loads to the ERCOT System. Sufficient transmission capacity plays a key role in ensuring system reliability while meeting the rapid growth in demand. While transmission still typically requires 3 to 5 years to be developed and energized, demand growth is escalating as new large loads are added to the ERCOT System faster (6 to 12 months) and in greater sizes than historical load interconnections.

As the new load interconnection timeline outpaces the development of transmission projects, the Texas Legislature recognized the need to update rules in the ERCOT Region. Previously, these rules did not allow ERCOT to include loads in planning studies unless they had executed an interconnection agreement. Updating these rules is essential to expedite the process in order to reliably serve the anticipated rapid demand growth. In December 2022, the Public Utility Commission of Texas (PUCT) amended 16 Texas Administrative Code (TAC) § 25.101 to implement Senate Bill (SB) 1281 (87th

Legislature). The amendments included the requirement for any review conducted by ERCOT to incorporate historical load, forecasted load growth, and additional load currently seeking interconnection. In June 2023, House Bill (HB) 5066 (88th Legislature) was enacted, introducing a new requirement to consider loads for which the electric utility has yet to execute an interconnection agreement, as determined by the electric utility responsible for serving the load.

These new requirements combined with unprecedented economic growth led to a significant increase in large loads considered in the 2024 RTP (i.e., data centers, hydrogen and hydrogen-related manufacturing, cryptomining, and electrification). The forecasted summer peak demand for 2030 exceeded 150 GW, of which approximately 50 GW is large load. Close to 40 GW of that forecasted demand for large loads seeking interconnections is due to the new rules permitting additional forms of Substantiated Load beyond just load with an executed interconnection agreement.

While the new rules allow for the consideration of more load in the planning processes, the policy governing the inclusion of planned Generation Resources in the planning models remains unchanged. Currently, planned Generation Resources can only be added to the planning models when they meet the requirements established in paragraph (1) of the ERCOT Planning Guide Section 6.9, *Addition of Proposed Generation to the Planning Models*. Under this rule, the total available generation was less than 120 GW in the 2030 summer peak case, based on the 2024 RTP renewable and battery dispatch assumptions. This indicated a supply deficit of more than 30 GW to meet the demand. Additional steps outside of the current planning practices were needed to create planning models capable of accommodating the unprecedented load growth. To achieve a balance between supply and demand, ERCOT incorporated generators that have not yet met the Planning Guide Section 6.9 requirements into the planning models and presented the assumptions¹ at the May 2024 Regional Planning Group (RPG) meeting. ERCOT is currently developing a Planning Guide Revision Request (PGRR) to adjust the criteria for adding Generation Resources to the planning models and will initiate stakeholder discussions on this topic in 2025.

In addition to the magnitude of the load growth, the “flat” or constant nature of the large loads adds another layer of complexity. Most of the large loads seeking interconnection are expected to maintain consistent electricity consumption regardless of season, day, or time. Ensuring reliable interconnection for these loads requires a resource mix capable of supporting their 24/7 demand. According to the capacity expansion results from the 2024 ERCOT Long-Term System Assessment (LTSA) high large load adoption scenario, this resource mix requires additional dispatchable Generation Resources.

The new load forecast in the ERCOT Region creates significant investment opportunities for balanced generation growth. The right resource mix needs to be carefully considered to address not only the magnitude of the load growth but also its unique characteristics.

This unprecedented load growth prompted discussions about introducing a 765-kV backbone infrastructure to the ERCOT Transmission Grid. This 765-kV backbone would function as a

¹ https://www.ercot.com/files/docs/2024/05/14/2024_RTP_Update_May_2024_RPG.pdf.

superhighway, enabling current to flow more efficiently through long-distance transmission from resource-rich regions to load centers. The benefits of higher voltage transmission may include:

- Increased transfer capability to load centers;
- Flexibility on Generation Resource siting;
- Outage coordination capacity;
- Reduced impact to Texas consumers due to fewer Right of Way (ROW) requirements;
- Lower line losses;
- Possible retirement of series compensation devices; and
- Potential exit strategy for some current Generic Transmission Constraints (GTCs).

In July 2024, ERCOT filed the Permian Basin Reliability Plan Study² with PUCT, which included both a plan with traditional 345-kV import paths and a plan with 765-kV import paths to the Permian Basin area. The PUCT approved the Permian Basin Reliability Plan in October 2024, irrespective of the voltage level, and is anticipated to make a determination on the import paths voltage level by May 1, 2025. To facilitate the PUCT's discussion, the 2024 RTP developed two plans to address statewide reliability needs: the traditional 345-kV plan and the 765-kV plan. ERCOT intends to file the 345-kV vs 765-kV comparison plan based on the 2024 RTP together with the dynamic analysis and congestion analysis with PUCT in January 2025.

Unprecedented load growth, insufficient generation to meet this demand, and the faster pace of load interconnections compared to transmission and generation development moved the ERCOT Region into a new era of planning. These challenges are further compounded by ongoing changes in the resource mix. The locations of both load and generation are crucial in driving flow patterns and determining transmission needs. While the generation interconnection queue offers some insight into future generation development, the location of loads is less predictable, especially given the much shorter lead times for load interconnections. This adds complexity to transmission planning, as flow patterns and stressed import paths may change between planning cycles. Planning practices and policies must be reviewed to address these new challenges. ERCOT has been and will continue to work with stakeholders to identify the necessary changes in the planning processes to ensure system reliability.

The rapid growth of wind, solar, and energy storage resources, coupled with increased retirements of coal and gas generators, led to greater reliance on intermittent, renewable generation to meet the higher system demand. This resulted in more long-distance power transfers as renewable generation is typically located far from the load centers. Similar to the 2023 RTP, the 2024 RTP study results highlighted the ongoing need for major enhancements to the import pathways to both Central Texas and South Dallas. However, with a significant amount of large load incorporated in and around the

²*Reliability Plan for the Permian Basin under PURA § 39.167*, PUCT Project No. 55718, ERCOT Permian Basin Reliability Plan Study (July 25, 2024).

Corpus Christi area in the 2024 RTP, imports to Central Texas and South Dallas now primarily come from the Coast Weather Zone, specifically southwest of Houston, instead of from South Texas as indicated by the 2023 RTP study results. In addition to the import needs identified in the 2023 RTP for South Dallas, the 2024 RTP identified the need for four new 345-kV import paths and significant reactive power support due to the substantial increase of large load in the area. Detailed findings of the 2024 RTP reliability analysis can be found in section 3 of this report.

Overall, 274 reliability projects were identified in the 2024 RTP's 345-kV plan to address all reliability violations compared with 173 projects in the 2023 RTP, 89 projects in the 2022 RTP, 67 projects in the 2021 RTP, and 50 projects in the 2020 RTP. This trend emphasizes the transmission challenges associated with the rapidly changing grid.

The majority of planned improvements identified in the 2024 RTP 345-kV plan are 138-kV and 345-kV system upgrades. The projects identified as 345-kV consist of new substations, transmission line additions, upgrades and rebuilds, new 345/138-kV transformers, existing 345/138-kV transformer upgrades, and the addition of voltage support devices.

ERCOT identified the following noteworthy reliability projects in the 2024 RTP 345-kV plan:

- Roscoe 345-kV substation and 345/138-kV transformer additions in Nolan County
- White Baker 345-kV substation, 345/138-kV transformers, and White Baker to Bakersfield 345-kV double-circuit line additions in Pecos County
- Prong Moss 345/138-kV transformer additions in Howard County
- Faulkner and new 90005Tap 345/138-kV substations, 345/138-kV transformers, and Solstice to Faulkner to new 90005Tap 345-kV double-circuit line additions in Reeves and Pecos Counties
- Farmland to Longdraw 345-kV line upgrade in Lynx and Borden Counties
- Kiamichi Energy Facility 345-kV reactors addition in Pittsburg County
- Watermill to Tri Corner to Trinidad 345-kV double-circuit line upgrades in Dallas and Henderson Counties
- Knob Creek Switch to Salado Switch 345-kV line upgrade in Bell County
- Watermill Switch to Loop Nine Switch 345-kV line upgrade in Dallas County
- Bale to St. Johns Switch to Lake Creek SES 345-kV line upgrades in Falls and McLennan Counties
- Allen Switch to Pineforest POI 345-kV line upgrade in Collin and Hopkins Counties
- Royse area 345-kV line upgrades in Collin, Hopkins, Rockwall, Fannin, and Lamar Counties
- Lake Creek - Trading House Area 345-kV line upgrades in Hill and McLennan Counties
- Bell County East Switch to Salado Switch 345-kV line addition in Bell and Williamson Counties

- Seagoville Switch 345/138-kV transformer addition along with Tri Corner to Seagoville Switch 345-kV line upgrade in Dallas County
- Big Brown SES to Navarro 345-kV double-circuit line upgrades along with Navarro to Venus North 345-kV line upgrade in Freestone, Navarro, and Ellis Counties
- Cedar Hill Switch to Merrifield Road 345-kV line upgrade in Dallas County
- Navarro to Outlaw Switch 345-kV line upgrade in Navarro County
- Miller Road 345/138-kV substation addition and 345-kV line reterminations in Dallas and Ellis Counties
- Miller Road to Oystercatcher POI 345-kV line upgrade in Ellis County
- Bell County East Switch to Sandow Switch, Bell County East Switch to Littlepond, and Bell County East Switch to Brangus Switch 345-kV line upgrades in Milam and Bell Counties
- Tyler Grande to Kings Creek 2 345-kV line upgrade in Smith and Dallas Counties
- Watermill area 345-kV line additions and reactive support in Dallas, Hill, Leon, and Navarro Counties
- Limestone Plant to Jewett 345-kV line upgrade in Leon and Limestone Counties
- Tri Corner to Forney Switch 345-kV line upgrade in Dallas and Kaufman Counties
- Elkton (3105) to Tri Corner (2432) 345-kV line additions and upgrades in Dallas, Kaufman, Smith, and Van Zandt Counties
- Batchler Road to Watermill 345-kV double-circuit line upgrades in Dallas County
- Sam Switch to Four Brothers Switch 345-kV line upgrade in Hill County
- Mitchell Bend Switch to Carmichael Bend Switch to Benbrook Switch 345-kV line upgrades in Grayson, Hood, and Tarrant Counties
- Skyview 345/138-kV substation addition and Liggett Switch to Norwood Switch 345-kV line upgrade in Dallas and Tarrant Counties
- Seagoville to Greene Road 345-kV double-circuit line additions in Dallas County
- Benbrook Switch 345/138-kV transformer upgrades in Tarrant County
- Desoto Switch 345/138-kV transformer addition in Dallas County
- Carter 345/138-kV substation addition in Tarrant County
- Rocky Creek 345/138-kV transformer addition Tarrant County
- Greene Road to Wilmer 345-kV double-circuit line additions in Dallas County
- Forney to Seagoville Switch 345-kV line upgrade in Dallas and Kaufman Counties
- Gunter 345/138-kV substation addition in Grayson County
- Lavon Switch 345/138-kV substation expansion in Collin County
- Carter 345/138-kV substation addition in Tarrant County
- Gibson to Brangus Switch 345-kV line upgrade in Milam County

- Lytton Springs 345/138-kV transformer addition in Caldwell County
- Austrop 345/138-kV transformer addition in Travis County
- Lytton Springs to Garfield to Austrop 345-kV line upgrade in Caldwell, Bastrop, and Travis Counties
- Cachena substation 345-kV reactor addition in Lavaca County
- Dunlap 345/138-kV transformer addition in Travis County
- Grissom to Raptor 345-kV line upgrade in Bee and Goliad Counties
- Brownsville 345-kV line additions and upgrades in Cameron County
- Lon Hill area 345-kV line additions and upgrades in Nueces, San Patricio, and Bee Counties
- Coletto to Haber to Raptor 345-kV double-circuit additions in Victoria and Goliad Counties
- Nelson Sharpe 345/138-kV transformer addition in Nueces County
- Naismith 345/138-kV transformer addition in San Patricio County
- Limmer 345-kV capacitor bank addition in Williamson County
- Euclid 345-kV capacitor bank addition in Caldwell County
- Lytton to Fayette Power Plant to Winchester 345-kV line upgrades in Caldwell, Fayette, and Bastrop Counties
- Hillje to Zorn 345-kV line upgrades in Wharton, Fayette, Bastrop, Caldwell, and Guadalupe Counties
- Gilleland Creek 345-kV capacitor bank addition in Travis County
- Shaula to Elm Creek to Cachena 345-kV line upgrades in Wilson, Guadalupe, Dewitt, and Lavaca Counties
- Voss Lake 345/138-kV substation expansion and Bell County East to Voss Lake 345-kV double-circuit line upgrade in Milam County
- Muscovy 345/138-kV substation addition and Salado Switch to Muscovy to Voss Lake 345-kV double-circuit addition in Milam and Williamson Counties
- Holman to Fayette Power Plant 345-kV line upgrade in Fayette County
- Haber 345-kV capacitor bank additions in Victoria County
- Old Hickory 345-kV cut in substation and 345-kV line additions in Goliad and Jackson Counties
- Angstrom to Whitepoint 345-kV line upgrade in San Patricio County
- La Palma 345/138-kV transformer addition in Cameron County
- Amber substation 345-kV capacitor bank additions in Nueces County
- Fayette Power Plant to Salem 345-kV line upgrade in Fayette, Austin, and Washington Counties
- Static substation 345-kV reactor addition in Refugio County

- Zoisite and Radiant substations 345-kV capacitor bank additions in Matagorda and Jackson Counties
- South Texas Project to Radiant to Cobra to Haber 345-kV line upgrade in Matagorda and Jackson Counties
- Radiant to Cobra to Haber 345-kV double-circuit line addition in Jackson County
- Hillje to South Texas Project 345-kV line upgrades in Wharton and Matagorda Counties
- South Lane City 345/138-kV substation expansion in Wharton County
- Cedar Bayou 345/138-kV transformer additions and upgrades in Chambers County
- Shamburger Switch to Elkton 345-kV double-circuit line addition in Smith County
- TNP One Plant 345/138-kV substation expansion and 345/138-kV transformer additions in Brazos and Robertson Counties
- Monticello SES to Cartwheel POI 345-kV line upgrades in Titus and Hopkins Counties
- Stryker Creek SES to Trinidad SES 345-kV line upgrade in Cherokee and Henderson Counties
- Trinidad SES to Richland Chambers 345-kV line upgrade in Henderson and Freestone Counties
- Umbra to Monticello SES 345-kV line upgrade in Franklin and Titus Counties
- Umbra to Woodard Switch 345-kV line upgrade in Franklin and Lamar Counties
- Thorn Tree Switch to Saltillo Switch 345-kV line upgrade in Titus and Hopkins Counties
- Big Brown SES West to Jewett 345-kV line upgrade in Freestone and Leon Counties
- Big Brown SES East to Jewett 345-kV line upgrade in Freestone and Leon Counties

The 2024 RTP includes an economic assessment of the ERCOT Transmission Grid for years 2026 and 2029 using both the production cost savings test and the generator revenue reduction test. The 2024 RTP economic study used the 2023 RTP final reliability cases as the start case to develop the 2026 and 2029 study cases. It did not incorporate the additional large loads included in the 2024 RTP reliability cases because the reliability projects (345-kV plan or 765-kV plan) needed to reliably supply such a significant amount of large load additions were not available at that time. Through this economic assessment, ERCOT identified transmission congestion and tested various transmission improvements to address this congestion in a cost-effective manner (as defined by ERCOT's economic planning criteria). Eight economic transmission improvement projects were evaluated in the 2024 RTP. In addition to the production cost savings test and the generator revenue reduction test, the consumer energy cost reduction test results were also provided for informational purposes only at this time. Note that Nodal Protocol Revision Request (NPRR) 1247,³ which adds the consumer energy cost reduction test (also known as the congestion cost savings test) to Protocols Section 3.11.2,

³ <https://www.ercot.com/mktrules/issues/NPRR1247#keydocs>.

Planning Criteria, was recommended for approval by the ERCOT Board of Directors in December 2024. Detailed economic analysis results can be found in section 4 of this report.

The estimated project completion years provided in the 2024 RTP report were selected to address reliability needs in a timely manner. While TOs are expected to meet these project completion dates, actual project timelines may vary due to factors such as construction clearance, regulatory or governmental approval times, equipment availability, land acquisition, and resource constraints.

The transmission projects identified in the RTP do not represent ERCOT's endorsement of those projects. Instead, they represent suggested CAPs for the reliability criteria violations identified or economic projects with sufficient economic benefits produced under the system conditions studied in the RTP. The scopes of projects identified in the RTP may change based on further analysis by ERCOT or the TPs that indicate better alternatives or a need to modify the projects due to changes in expected generation, load forecasts, or other system conditions. To confirm need, TPs should perform studies with the latest system conditions and develop applicable reliability projects to resolve any reliability criteria violations and economic projects where sufficient economic benefits to pass the specified test can be demonstrated.

For projects that are subject to ERCOT Protocols Section 3.11.4, *Regional Planning Group Project Review Process*, a review shall be conducted in accordance with the process described therein. For a project that is under RPG review when the RTP is developed, a placeholder project will be used if the need is identified. Projects requiring RPG endorsement will be reviewed in future assessments (where sufficient lead-time exists), such as future RTPs, to ensure the identified system facilities are still needed.

The TOs will provide ERCOT with additional details on project scope, project cost, and an implementation schedule with completion date(s) for each identified project. This information from the TOs may be provided through further RPG review and/or Transmission Project Information Tracking (TPIT) updates in accordance with ERCOT Planning Guide Section 6.4.1, *Transmission Project and Information Tracking Report*.

Table of Contents

Executive Summary	i
1. 2024 Regional Transmission Plan.....	1
1.1. Stakeholder Involvement.....	1
1.2. Standards and Regulations	1
1.3. Confidentiality and Report Posting.....	2
2. 2024 Regional Transmission Plan Process.....	3
2.1. Permian Basin Load Forecast and Large Load Additions	4
2.2. Adoption of the ERCOT Rooftop Solar Growth and Electrical Vehicle (EV) Load Impact Forecast	7
2.3. Generation Assumptions	8
2.4. Reliability Analysis.....	9
2.5. Economic Analysis	12
3. Findings from Reliability Analysis	13
3.1. Reliability Projects and Constraint Management Plans for the 345-kV Plan	13
3.1.1 West Texas Study Findings.....	15
3.1.2 Central Texas Study Findings	16
3.1.3 South Dallas Study Findings	19
3.1.4 Generation Changes in the Coast Weather Zone	21
3.1.5 Other Findings	21
3.2. 765-kV Plan Findings	22
3.3. Sensitivity Analysis	24
3.4. Short Circuit Analysis	25
3.5. Long Lead Time Equipment Analysis.....	25
4. Economic Analysis.....	26
4.1. Study Assumptions and Methodology	30
4.2. Top Constraints for 2026 and 2029 Study Years.....	30
4.3. Projects Selected for Evaluation	32
4.4. Project Evaluation Results	33
5. Appendices	34

1. 2024 Regional Transmission Plan

This report documents the 2024 Regional Transmission Plan (RTP) assessment performed by ERCOT Grid Planning. It is intended, in part, to satisfy ERCOT's requirements under North American Electric Reliability Corporation (NERC) Reliability Standards, ERCOT Protocols Section 3.11, and ERCOT Planning Guide Sections 3 and 4.

The RTP study is conducted annually for the entire ERCOT System. The 2024 RTP's near-term and long-term planning horizon analysis evaluated the reliability needs of the ERCOT transmission system for the years 2026 through 2030. As required by NERC Reliability Standard TPL-001-5.1, the 2024 RTP included a steady-state analysis of summer peak conditions for years 2026 (year 2), 2027 (year 3), and 2029 (year 5); and off-peak conditions for 2027 (year 3); and a short-circuit analysis of summer peak conditions for 2027 (year 3). The 2024 RTP also included steady-state analysis of summer peak conditions for 2030 (year 6), representing the long-term planning horizon. Year six, *i.e.* 2030, was selected based on the rationale that most transmission upgrades in the ERCOT region can be completed within five to six years from the date when the need is identified. In addition to analyzing the reliability needs of the system, the 2024 RTP also evaluated economic/efficiency needs of the ERCOT system for years 2026 and 2029.

1.1. Stakeholder Involvement

The development of the RTP is a collaborative process. ERCOT worked with NERC-registered Transmission Planners (TPs), Transmission Owners (TOs), and other stakeholders to develop the input assumptions and scope of technical studies that define the 2024 RTP. These assumptions are described in the RTP Scope and Process document and were presented to the stakeholder community at Regional Planning Group (RPG) meetings. Stakeholders were provided with routine updates on the input assumptions and supporting analysis performed for the 2024 RTP in RPG meetings. Feedback and comments from the RPG were incorporated into the RTP Scope and Process document.

The RPG is responsible for reviewing and providing comments on proposed transmission projects in the ERCOT Region. Under ERCOT Protocols Section 3.11.3, participation in the RPG is required of all Transmission Service Providers and is open to all Market Participants, consumers, other stakeholders, and Public Utility Commission of Texas (PUCT) Staff.

ERCOT worked with TPs, TOs, and other stakeholders to study the existing system and identify system upgrades and new transmission projects to ensure continued system reliability.

1.2. Standards and Regulations

The RTP assessment was conducted based on requirements in NERC Reliability Standards, ERCOT Protocols, and the ERCOT Planning Guide.

ERCOT performed its steady-state reliability assessment in accordance with NERC Reliability Standard TPL-001-5.1, Transmission System Planning Performance Requirements. A portion of the RTP assessment also addressed some requirements in NERC Reliability Standards FAC-002⁴ and IRO-017.⁵

ERCOT Protocols Section 3.10.8.4(3) requires ERCOT to identify additional Transmission Elements that have a high probability of providing significant added economic efficiency to the ERCOT market through the use of Dynamic Ratings and request such Dynamic Ratings from the associated ERCOT Transmission Service Provider (TSP). This report identifies such Transmission Elements as part of its economic analysis.

The RTP assessment adheres to ERCOT Planning Guide Section 3.1.1.2, which provides guidelines regarding completion of the RTP. This section requires that ERCOT complete and publish the final RTP report no later than December 31 each year. Additionally, ERCOT Planning Guide Section 4 and ERCOT Protocols Section 3.11.2 specify the transmission planning criteria to be used in the RTP assessment.

1.3. Confidentiality and Report Posting

The RTP report is shared with internal and external stakeholders. One redacted version of the RTP is created by removing, at a minimum, any confidential data such as the list of long lead-time equipment. This report is shared with ERCOT stakeholders via the MIS Secure area. A public version of the RTP report is also created by removing, at a minimum, any confidential data and ERCOT Critical Energy Infrastructure Information (ECEII). This report is posted to the ERCOT website.

⁴ FAC-002, Requirement R4.

⁵ IRO-017, Requirements R3 and R4.

2. 2024 Regional Transmission Plan Process

The RTP study process is illustrated in Figure 1. The initial start cases for the reliability analysis were prepared during the case conditioning stage. For the 2024 RTP, this case conditioning stage included the use of the “bounded higher-of” methodology to determine appropriate weather zone load levels for the RTP study. The details of this methodology can be found in ERCOT Planning Guide Section 3.1.7, *Steady State Transmission Planning Load Forecast*. In the 2024 RTP, the Permian Basin load forecast from S&P Global Commodity Insights (formerly IHS Markit) was utilized for the West and Far West Weather Zones. In addition, a significant number of large loads substantiated by signed contracts and officer letters from the Transmission Service Providers (TSPs) were incorporated in the 2024 RTP.

Following case conditioning, a reliability analysis was conducted on the base cases to determine the Corrective Action Plans (CAPs) needed to meet ERCOT and NERC reliability requirements. In addition to the base cases, the 2024 RTP included sensitivity cases, a short-circuit analysis, a cascade analysis, a known outages study, and a multiple element outage analysis as required by NERC Reliability Standard TPL-001-5.1. A minimum deliverability analysis was performed based on the criteria defined in ERCOT Planning Guide Section 4.1.1.7, *Minimum Deliverability Criteria*, and the threshold approved by the ERCOT Board of Directors.⁶ Additionally, a maintenance outage analysis was performed based on the criteria defined in ERCOT Planning Guide Section 4.1.1.8, *Maintenance Outage Reliability Criteria*. Economic analysis was also conducted to identify transmission projects that allow reliability criteria to be met at a lower total cost. The detailed scope, process, and input assumptions used in conducting reliability and economic analyses are available in Appendices A, B, and C.

⁶ https://www.ercot.com/files/docs/2022/06/28/Minimum_Deliverability_Criteria_Thresholds.pdf.

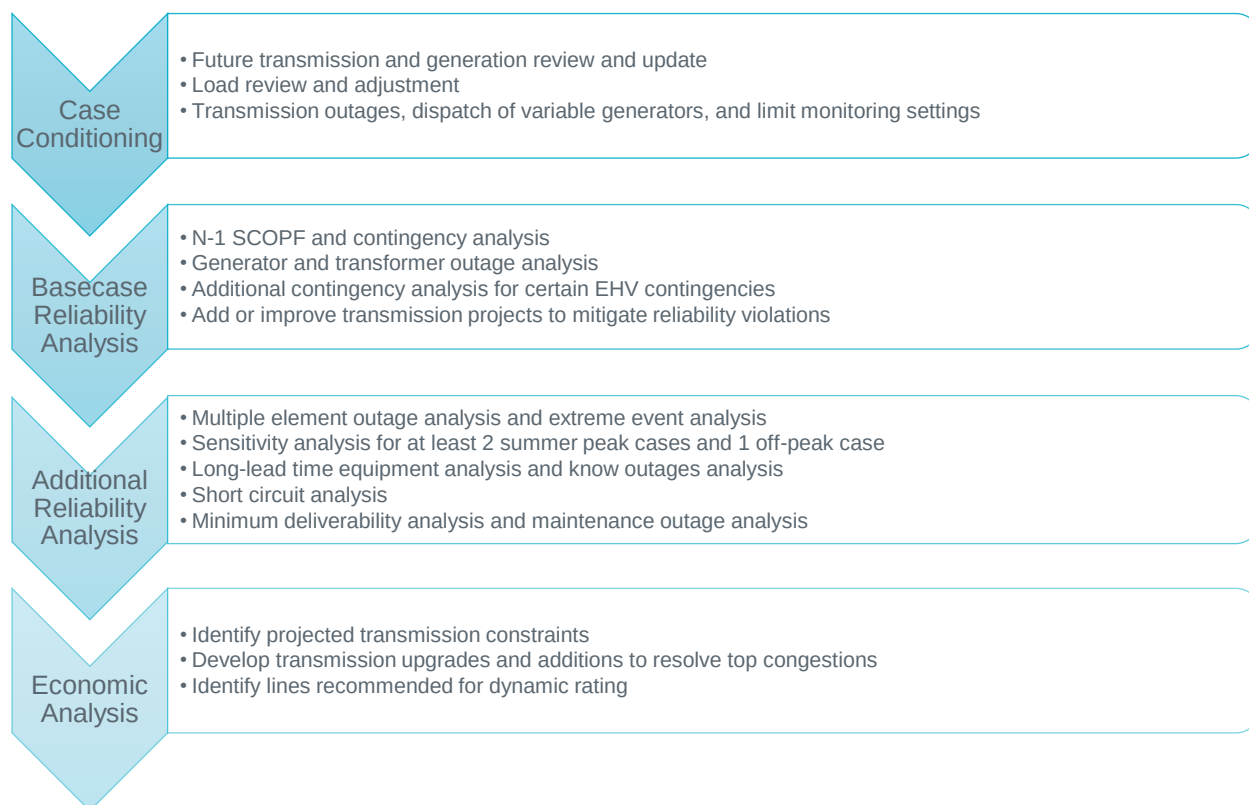


Figure 1: 2024 RTP Transmission Planning Process

ERCOT utilized the following software tools while performing the 2024 RTP:

- PSS/E version 35 was used to develop the conditioned cases;
- PowerWorld version 23 with Security Constrained Optimal Power Flow (SCOPF) and its SIMAUTO functionality were used to perform AC SCOPF analysis and to run generator and transformer outage analysis;
- PowerWorld version 23 was used to screen critical contingencies while evaluating P3 (generator outage) and P6-2 (transformer outage) planning events;
- PowerWorld version 23 was used to perform multiple element outage analysis and cascading analysis; and
- UPLAN version 12.3 was used to perform security-constrained economic analysis.

2.1. Permian Basin Load Forecast and Large Load Additions

To better prepare for the challenges in transmission planning introduced by the rapid load growth in the Permian Basin and the short lead time of oil and gas load interconnection requests, ERCOT and TSPs serving West Texas oil and gas load have been working proactively. This effort aims to better understand oil and gas activities and growth, positioning the Texas grid for potential long lead time transmission enhancements needed to reliably serve the fast-growing loads.

In 2019, ERCOT completed the Delaware Basin Load Integration Study with extensive input from TSPs, identifying a five-stage transmission upgrade road map to reliably serve different levels of Delaware Basin load. In 2021, ERCOT completed the Permian Basin Load Interconnection Study, informed by the TSP-sponsored IHS Markit study report for Permian Basin load forecast, which provided a detailed load forecast based on an in-depth analysis of oil and gas activities in the Permian Basin. In 2022, S&P Global Commodity Insights (formerly IHS Markit) refreshed the Permian Basin load forecast, showing continued load growth and grid connection needs. The load forecast from 2026 to 2040 in the Permian Basin area is shown in Figure 2. In July 2024, as directed by the PUCT, ERCOT developed the Permian Basin Reliability Plan which incorporated the S&P Global Commodity Insights Permian Basin load forecast. This plan contains both local upgrades and new import paths to address reliability issues in the Permian Basin area. The S&P Global Commodity Insights Permian Basin load forecast was also incorporated in the 2024 RTP.

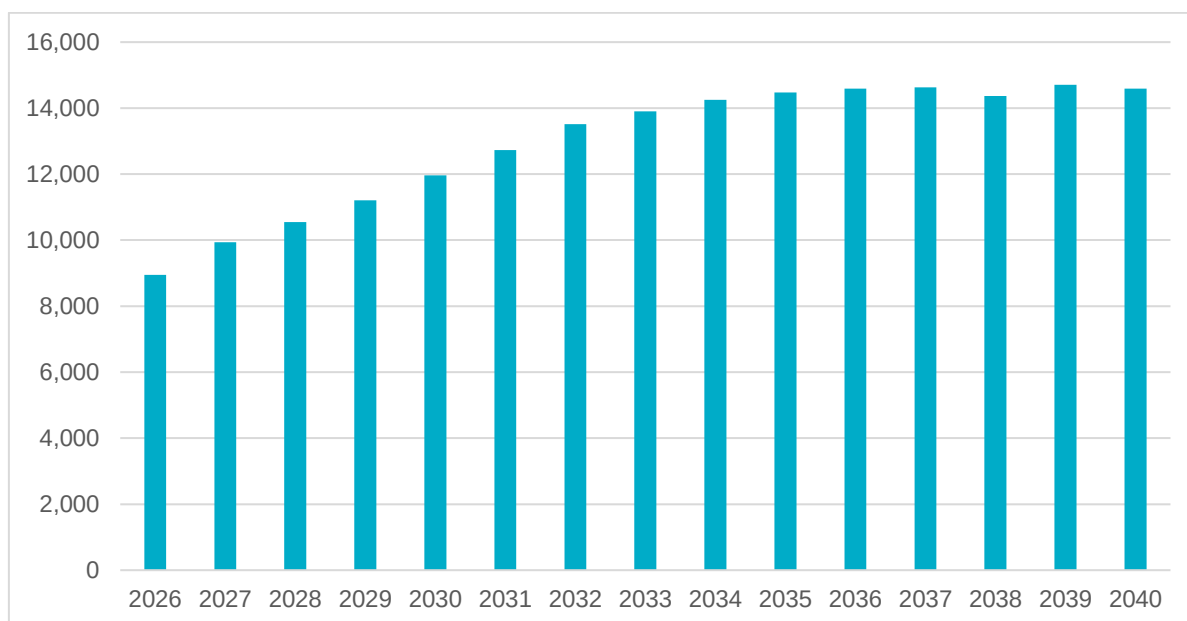


Figure 2: S&P Global Commodity Insights Permian Basin Summer Peak Load Forecast (MW)

The implementation of House Bill (HB) 5066 in the 2024 RTP brought substantial changes into the planning process, moving the ERCOT region into a new era of planning.⁷ Prior to the 2024 RTP, ERCOT only considered those large loads that had an executed contract with their electric utility to determine the appropriate load to include in the analysis. This resulted in the inclusion of approximately 8 GW of large loads in both the 2022 and 2023 RTPs. HB5066 introduced a new requirement to consider loads for which the electric utility has yet to sign an interconnection agreement, as determined by the electric utility responsible for serving the load. Consequently, ERCOT considered attestation

⁷ Tex. H.B. 5066, 88th Leg., R.S. (2023).

letters from TSP officers to determine the appropriate load to include. With both signed contracts and TSP officer letters considered, the large load incorporated in the 2024 RTP exceeded 50 GW.

Figure 3 shows the amount of large load included in each study year. These large loads include cryptocurrency load, data center load, and hydrogen and hydrogen-related manufacturing. The breakdown of different load types is shown in Figure 4.

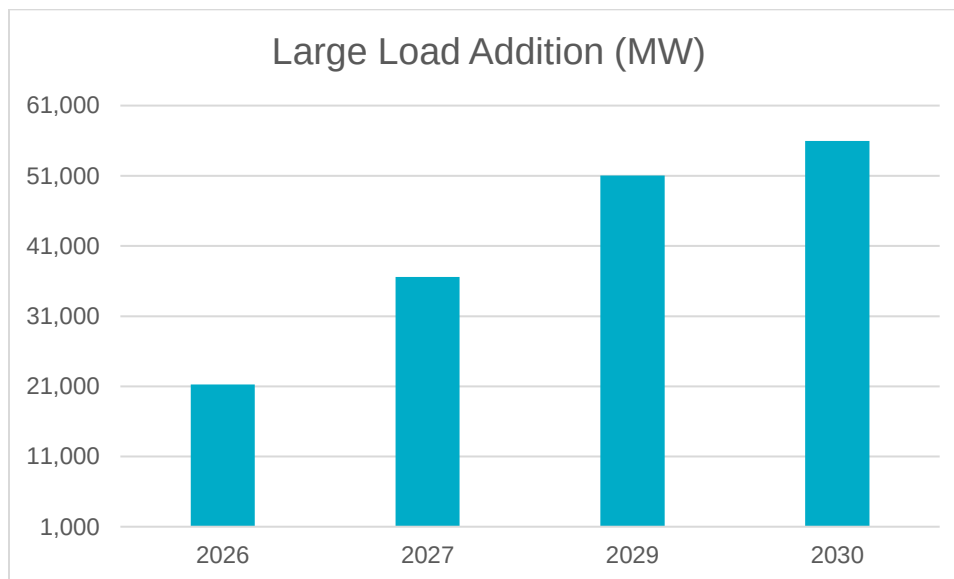


Figure 3: Large Load Addition in 2024 RTP (MW)

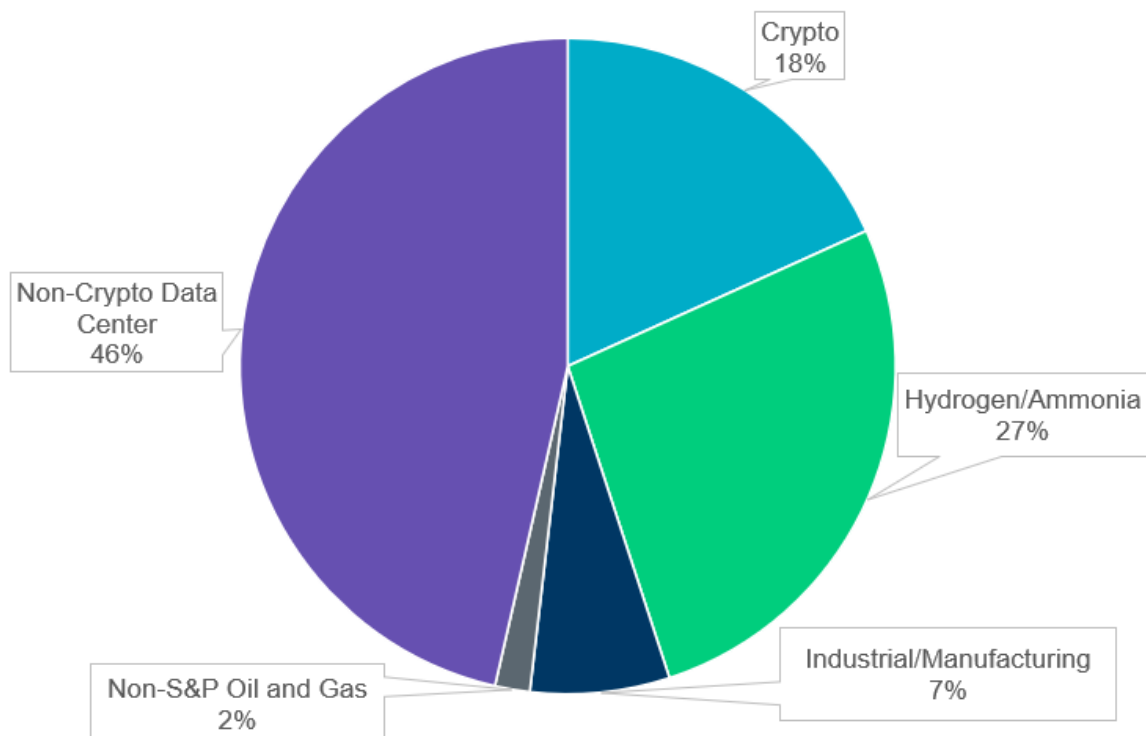


Figure 4: Types of Large Loads in 2024 RTP

Another critical characteristic associated with large load interconnection requests is the high concentration of large load requests in certain areas of the system. This created additional challenges in transmission planning including both import needs and local transmission needs. Figure 5 shows the location of the large loads in the 2024 RTP.

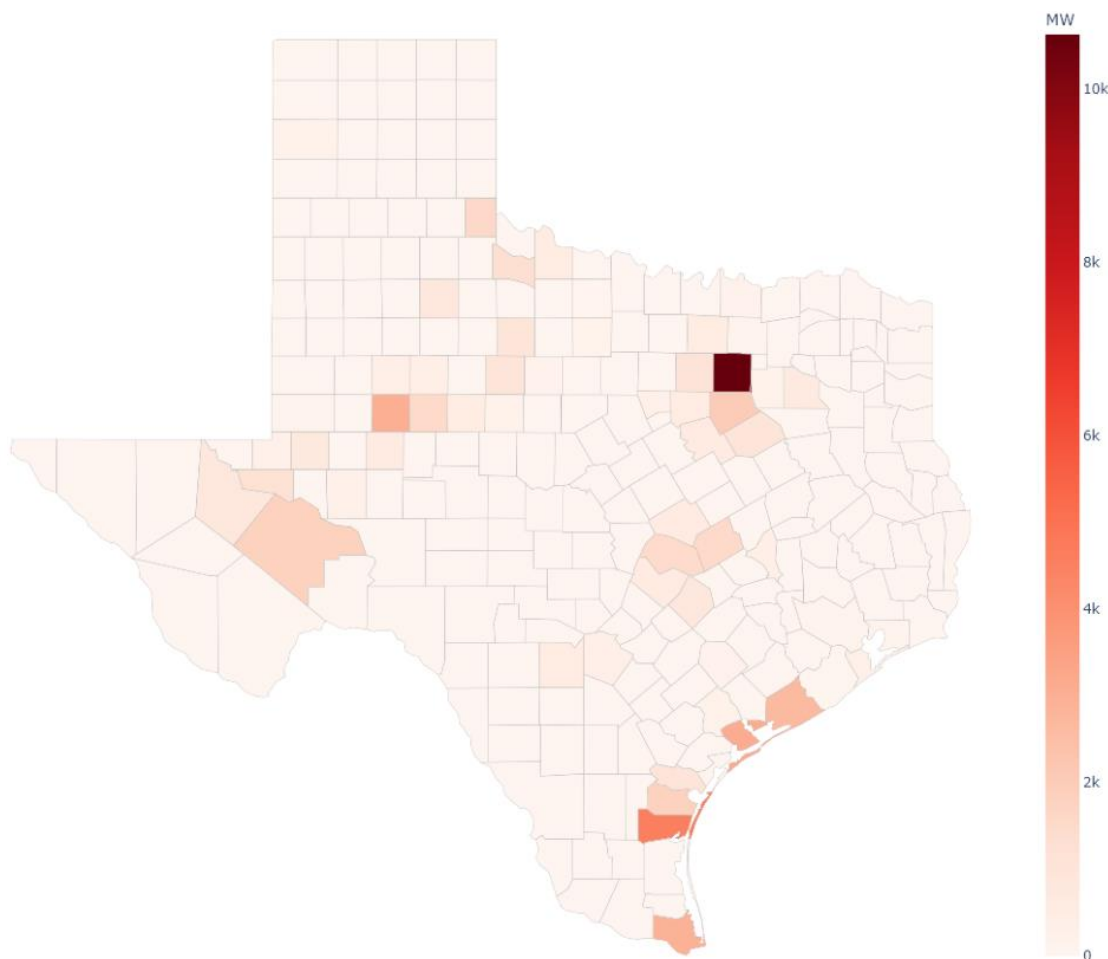


Figure 5: Location of Large Loads in 2024 RTP

2.2. Adoption of the ERCOT Rooftop Solar Growth and Electrical Vehicle (EV) Load Impact Forecast

The rapid growth in Distributed Generation (DG), especially in the solar photovoltaic less than 1 MW category, continued in the ERCOT region. The total DG at the end of 2023 is estimated to be more than 4,770 MW, as shown in Figure 6, which represents an approximately 24% increase from 2022.

Similar to the 2022 and 2023 RTPs, the impacts of the projected rooftop solar growth were incorporated as load reductions at the bus level in the 2024 RTP.

ERCOT Estimated Total DG Growth 2015-2023 (MW)

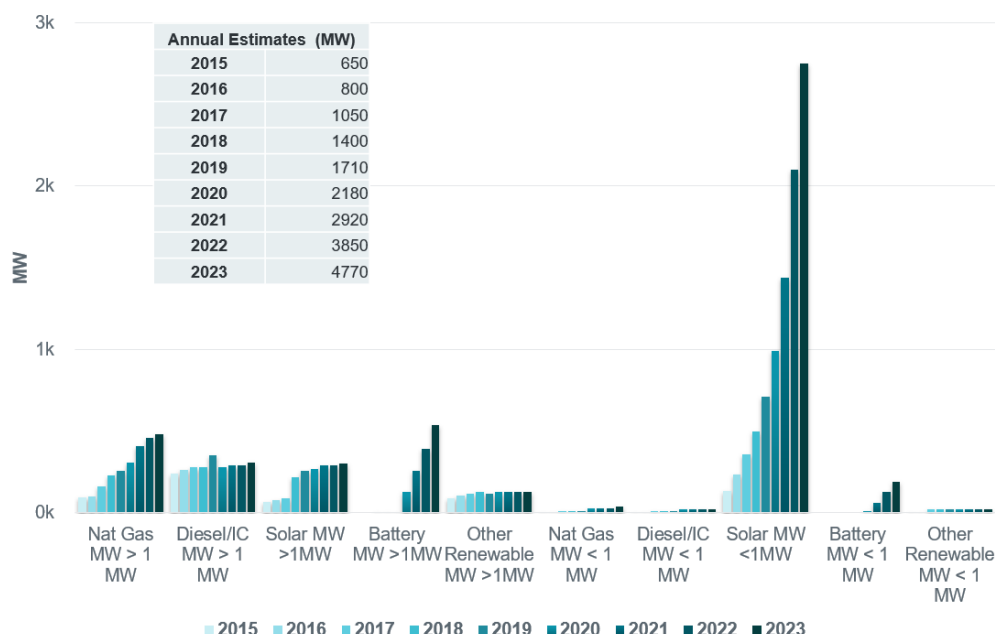


Figure 6: ERCOT Estimated Total DG Growth from 2015 to 2023 (MW)

Adoption of EVs is expected to increase significantly in the near future, which signifies a need to include EV load impacts in near-term planning studies. Similar to the 2023 RTP, the 2024 RTP incorporated the substation-level EV load impacts based on the methodology and process⁸ developed by the Brattle Group.

2.3. Generation Assumptions

To address the supply deficit needed to meet the forecasted demand before formal planning rules are established, ERCOT took the following steps outside of the normal planning process to incorporate more generation that have not met the Planning Guide Section 6.9 requirements yet into the planning models used in the 2024 RTP in the sequence below.

1. Added all units with a signed Interconnection Agreement (IA);
2. Added all dispatchable units with a Full Interconnection Study (FIS) officially underway (dispatchable units added to address 24/7 the nature of new loads); and
3. Sited approximately 11 GW of dispatchable generation, not in the interconnection queue, to resolve remaining generation shortfall in study cases and to address the 24/7 nature of the new loads.

The siting of the 11 GW of dispatchable generation was based on the following principles:

1. Focused on the largest system load pockets;

⁸ <https://www.ercot.com/files/docs/2023/08/28/ERCOT-EV-Adoption-Final-Report.pdf>.

2. Sought to utilize transmission capacity; and
3. Avoided non-attainment zones: counties designated by the Environmental Protection Agency (EPA) and Texas Commission on Environmental Quality (TCEQ) as non-attainment zones for ozone pollutants.

The location of the 11 GW dispatchable generation is shown in Figure 7.

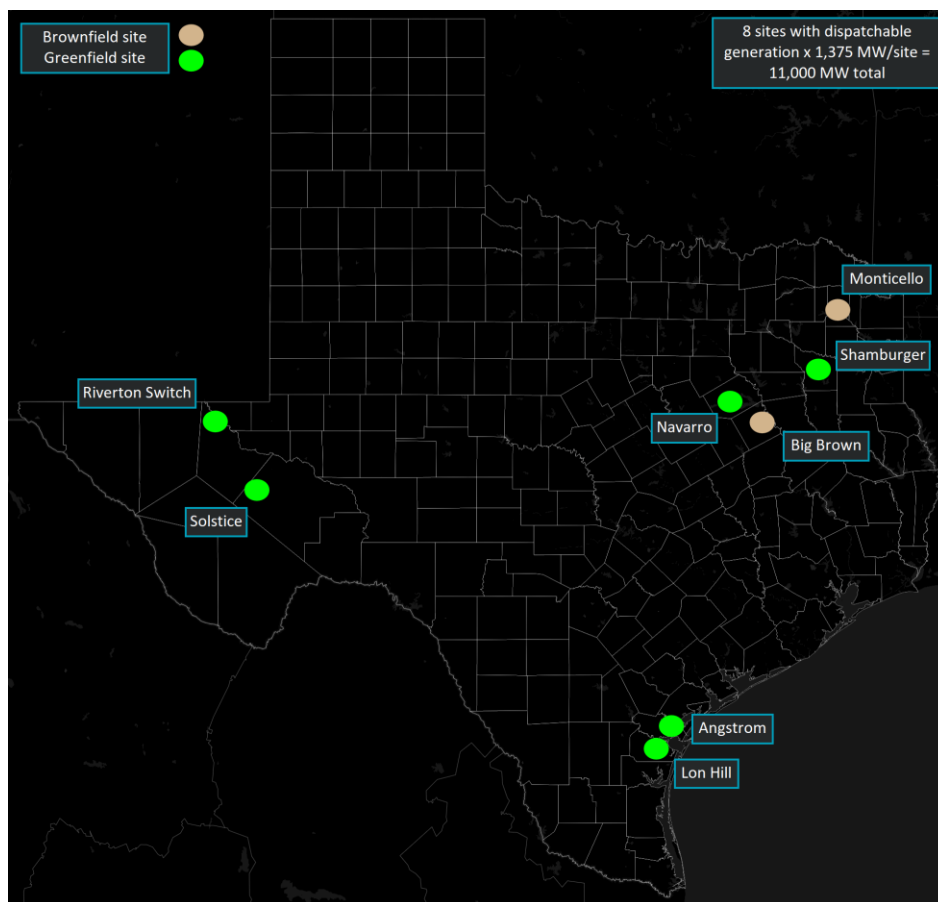


Figure 7: Location of Dispatchable Generation not in the Interconnection Queue

2.4. Reliability Analysis

The reliability analysis in the 2024 RTP was focused on the steady-state analysis requirements of NERC Reliability Standard TPL-001-5.1 and the ERCOT Planning Guide. The purpose of reliability analysis was to identify potential criteria violations and CAPs that may be used to resolve them. The RTP analysis included Security Constrained Optimal Power Flow (SCOPF) to identify unresolvable constraints. Loading and voltage levels at Bulk Electric System (BES) elements were monitored for all NERC planning events, including extreme events. ERCOT staff developed CAPs in collaboration with TPs to mitigate criteria violations in accordance with the NERC and ERCOT performance requirements.

The 2024 RTP reliability analysis included the following studies:

- SCOPF: Security Constrained Optimal Power Flow (SCOPF) was used to perform basic power flow and Contingency Analysis for P0, P1, and P2-1 planning events, and common tower outages as defined in ERCOT Planning Guide, Section 4.1.1.1, *Planning Assumptions*. SCOPF used generation cost data and other system constraints to give an optimal generation dispatch and unit commitment while maintaining the reliability of the system. In this analysis, the software simulated the removal of all elements of the Protection System and other automatic controls following the contingency event.
- Contingency Analysis: Basic contingency analysis routines in the power flow software were used to test P2-2, P2-3, P2-4, P4, and P5 planning events and extreme events.
- Multiple Element Contingency Analysis: Planning events P3 and P6 involve a first- and second-level contingency analysis. Such events were tested using multiple element contingency analysis. During this analysis, loss of elements due to the first contingency was followed by acceptable system adjustments before testing the effect of the second contingency event. The list of acceptable system adjustments included system reconfiguration, changes in voltage schedule, and re-dispatch of generation. Other contingency events such as P4 and P5 planning events and extreme events, which involved simultaneous removal of multiple elements, were also analyzed. Extreme events associated with the disruption of gas pipelines were also included.
- Cascading Analysis: Cascading analysis was conducted to test all planning and extreme events where a facility may be loaded above its relay loadability rating before mitigation measures can be taken. In this analysis, the software simulated the removal of all elements of Protection System and other automatic controls following the contingency event. This included tripping of generators and transmission elements which were loaded beyond their relay loadability limits. These contingencies were screened to detect potential cascade events for more detailed analysis.
- Short Circuit Analysis: In accordance with the agreement between ERCOT and TPs in the ERCOT region as required by NERC Reliability Standard TPL-001-5.1, Requirement R7 (revised in May 2020), ERCOT performed the short-circuit analysis to determine short-circuit currents for Resource Entity (RE)-owned facilities. The results of the short-circuit analysis included the magnitude of short-circuit current and the source impedance associated with each fault. These results were communicated to the NERC-registered Generator Owners (GOs). GOs completed a review of study results, acknowledged the findings, and provided a list of over-dutied circuit breakers and CAPs. In addition, GOs also confirmed the continued validity and implementation status of the facilities identified in the previous RTP.
- Long Lead Time Equipment Analysis: Under Requirement 2.1.5 of NERC Reliability Standard TPL-001-5.1, the impact of the possible unavailability of major transmission equipment with a lead time of one year or more was studied. The studies were performed with an initial condition of the identified long lead time equipment modeled as out of service, followed by P0, P1, and

P2 contingency events. The list of long lead time equipment was developed based on feedback from the TOs. The results of this analysis were communicated to the TOs.

- Sensitivity Analysis: ERCOT selected the summer peak conditions of 2026 and 2029 and off-peak conditions of 2027 for sensitivity analyses as required by Requirement 2.1.3 of NERC Reliability Standard TPL-001-5.1. ERCOT prepared the following sensitivity cases by varying the generation and load input assumptions:
 - The large loads substantiated by TSP officer letters were reduced by 50%.
 - The dispatchable generators that are not in the interconnection queue were turned off.
 The sensitivity analyses were performed using the same start cases as the 2024 RTP base case analysis.
- Known Outages Impact Analysis: Under Requirement 2.1.4 of NERC Reliability Standard TPL-001-5.1, the impact of known outages of generation or transmission facilities planned in the near-term planning horizon was studied. ERCOT issued Market Notices to collect known outages information from both TOs and GOs. TOs were required to provide technical rationales for their known outages selection for each study case included in the 2024 RTP. ERCOT developed the technical rationale⁹ for the selection of GO-submitted outages to be included in the corresponding study cases. The known outages from the TOs and the GO outages selected based on ERCOT-developed technical rationale were then used to study their impact on system performance under P0 and P1 contingencies.
- Minimum Deliverability Analysis: As required by ERCOT Planning Guide Section 4.1.1.7, Minimum Deliverability Criteria, ERCOT performed analysis to ensure the deliverability of 100% of capacity of Generation Resources, utilizing combined cycle, steam turbine, combustion turbine, hydro, or reciprocating engine technology, and for any Energy Storage Resource (ESR) with a duration greater than or equal to two hours. For ESRs with a duration less than 2 hours, a prorated deliverability was ensured. CAPs were proposed to address any reliability violations under the contingencies defined for the minimum deliverability criteria.
- Maintenance Outage Analysis: Pursuant to ERCOT Planning Guide Section 4.1.1.8, Maintenance Outage Reliability Criteria, ERCOT performed analysis to identify transmission solutions to address reliability issues under maintenance outage conditions during shoulder seasons. ERCOT created a 2030 fall peak case for the maintenance outage analysis to ensure sufficient outage window for equipment owners to perform necessary maintenance.

2.4.1 CAP Development

Under the ERCOT Planning Guide, reliability projects are those system improvements (projects) that are needed to meet NERC Reliability Standards or ERCOT planning criteria, which could not otherwise be met by simultaneously feasible, security-constrained re-dispatch of existing and planned generation. To develop this list of projects, grid simulation software was utilized which included the removal of all protection system elements and other automatic controls following the simulated

⁹ https://www.ercot.com/files/docs/2022/03/09/2022_RTP_TPL_001-5_Known_Outages_March_2022_RPG.pdf.

contingency events. These elements included devices designed to provide steady-state control of electrical system quantities, such as on-load tap-changing transformers, phase-shifting transformers, and switched capacitors and reactors.

A list of potential CAPs, or reliability projects, along with the corresponding limiting elements and contingencies, was communicated to the appropriate TP and/or TO. TPs and TOs reviewed the initial list of reliability-driven projects for their technical feasibility and estimated year of completion (considering necessary lead times). In some cases, the TOs also provided project alternatives. In instances where it is not feasible to construct a project prior to the identified date of need, ERCOT designed CMPs to mitigate the criteria violations until the permanent CAP can be put in-service. These mitigation actions were developed in collaboration with TPs and further communicated to ERCOT Operations. The results were posted on the ERCOT MIS Secure Area. Study findings were presented to stakeholders at regularly scheduled RPG meetings to solicit comments and suggestions.

2.5. Economic Analysis

ERCOT conducted an economic analysis to identify system improvements that allow ERCOT to meet NERC Reliability Standards and ERCOT planning criteria more economically than the continued dispatch of higher cost generation.

To identify such economically driven projects, ERCOT created a model for years 2026 and 2029. Details on the models developed for the 2024 RTP can be found in Appendices D and E.

According to the economic planning criteria described in paragraph (5) of ERCOT Nodal Protocols Section 3.11.2, ERCOT recommends an economic project if the annual production cost savings (PCS) exceed the first-year annual revenue requirement for the project. Based on the recent review of current market conditions, the first-year annual revenue requirement for a project was determined to be 12.9% of the estimated project cost.

In addition, ERCOT recommends an economic project if the annual generator revenue reduction (GRR) exceeds the average of the first three-year annual revenue requirement for the project, as allowed by the 16 Texas Administrative Code (TAC) § 25.101(b)(3)(A)(i)(I)(-b-), until the congestion cost savings (CCS) test becomes effective. Based on the recent review of current market conditions, the average of the first three-year annual revenue requirement for a project was 12.6% of the estimated project cost.

ERCOT submitted Nodal Protocol Revision Request (NPRR) 1247 in 2024 to add the new congestion cost savings test to the ERCOT Protocols, which is measured by the total consumer energy cost reduction resulting from a proposed project. NPRR1247 was recommended for approval by the ERCOT Board of Directors in December 2024 and is anticipated to be considered by PUCT for approval in January 2025. The results of the congestion cost savings test as proposed by NPRR1247 were also provided in this report for informational purposes at this time.

3. Findings from Reliability Analysis

3.1. Reliability Projects and Constraint Management Plans for the 345-kV Plan

The primary purpose of the 2024 RTP reliability analysis was to identify reliability criteria violations and potential CAPs to resolve them. Overall, the base reliability analysis identified a need for 274 CAPs. The detailed list of criteria violations and resulting CAPs can be found in Appendix E. Figure 8 illustrates the geographic location of the identified CAPs. The legend linking reliability projects and their associated map indices can be found in Appendix F.

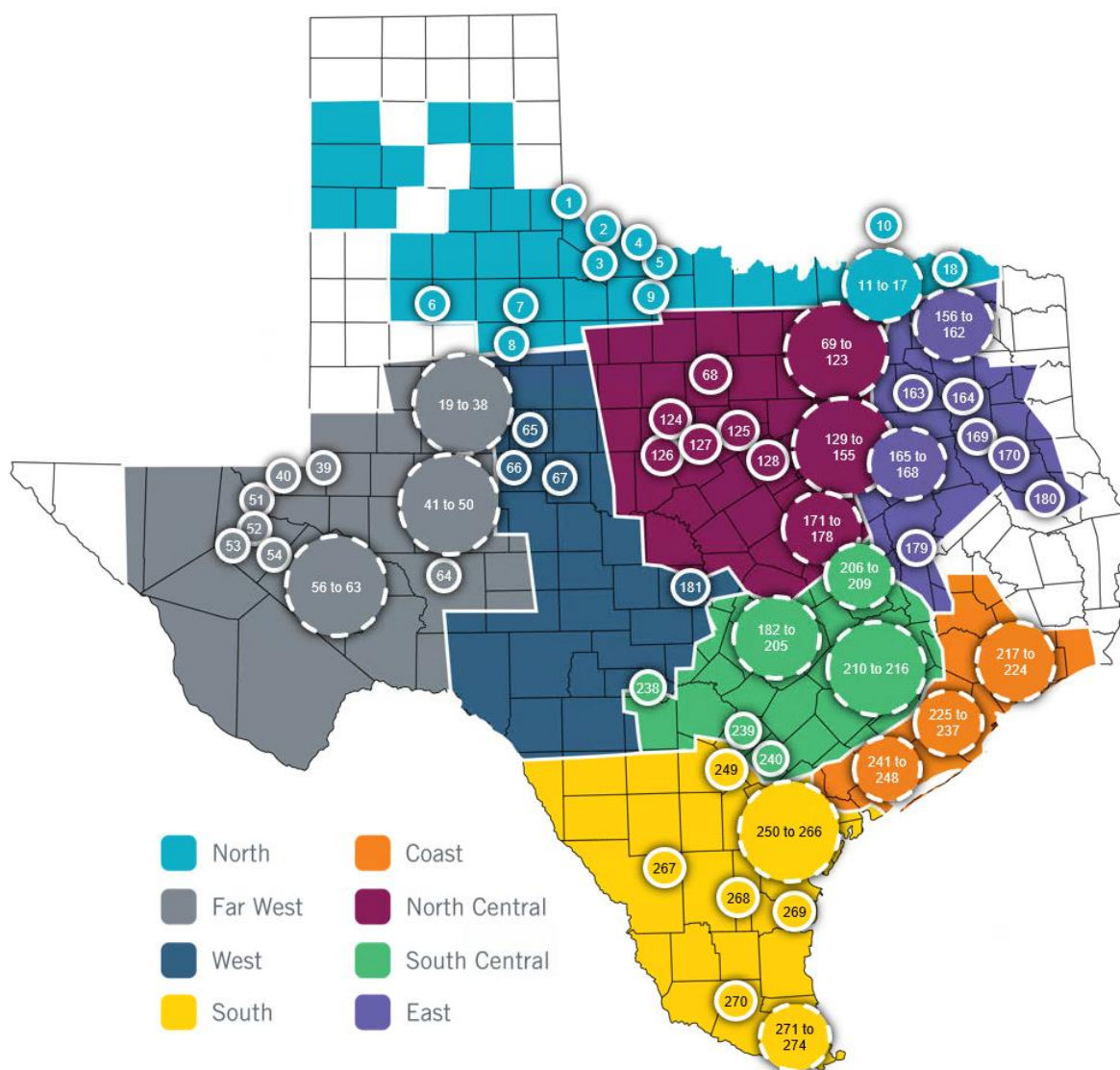


Figure 8: Geographic Locations of CAPs Identified in the 2024 RTP

Figures 9,¹⁰ 10, and 11 summarize the types of projects, their geographic locations, and associated voltage levels. Figure 12 distinguishes between projects that were newly identified in the 2024 RTP and projects that were identified in previous ERCOT planning studies or TSP studies.

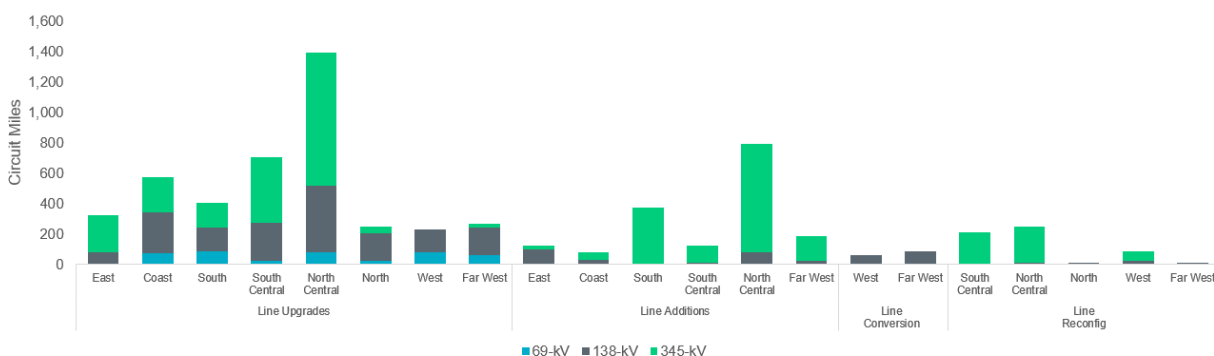


Figure 9: 2024 RTP Transmission Line Project Types by Weather Zone and Voltage Level

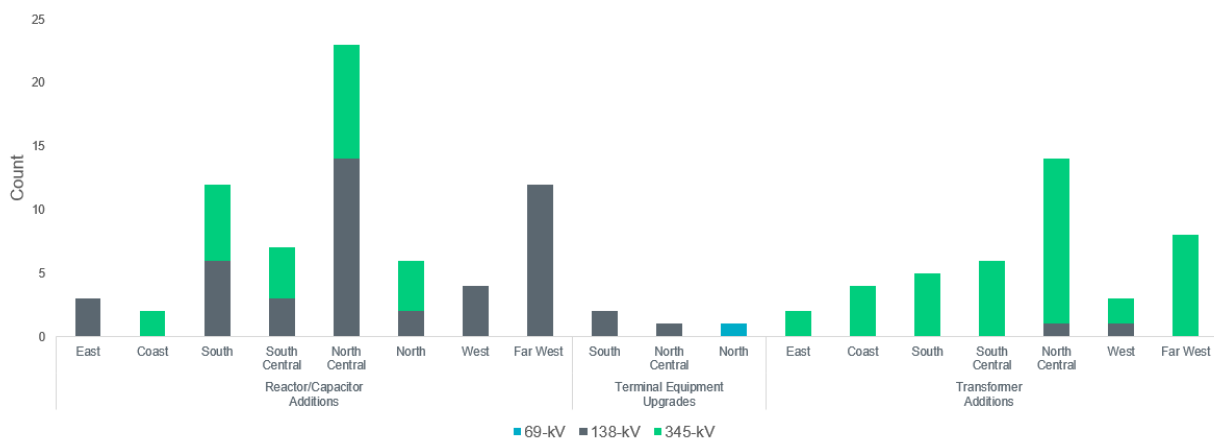


Figure 10: 2024 RTP Other Upgrades and Additions by Weather Zone and Voltage Level

¹⁰The 69-kV to 138-kV line conversion was included in the 138-kV category

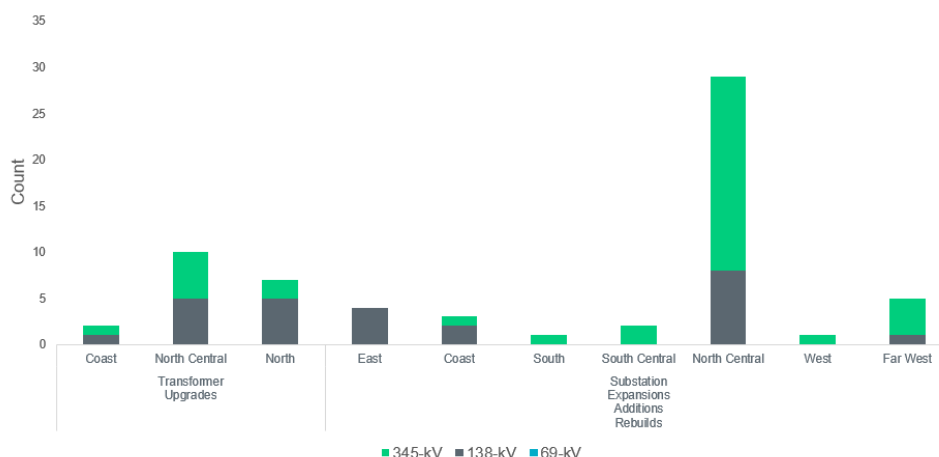


Figure 11: 2024 RTP Other Upgrades and Additions by Weather Zone and Voltage Level cont.

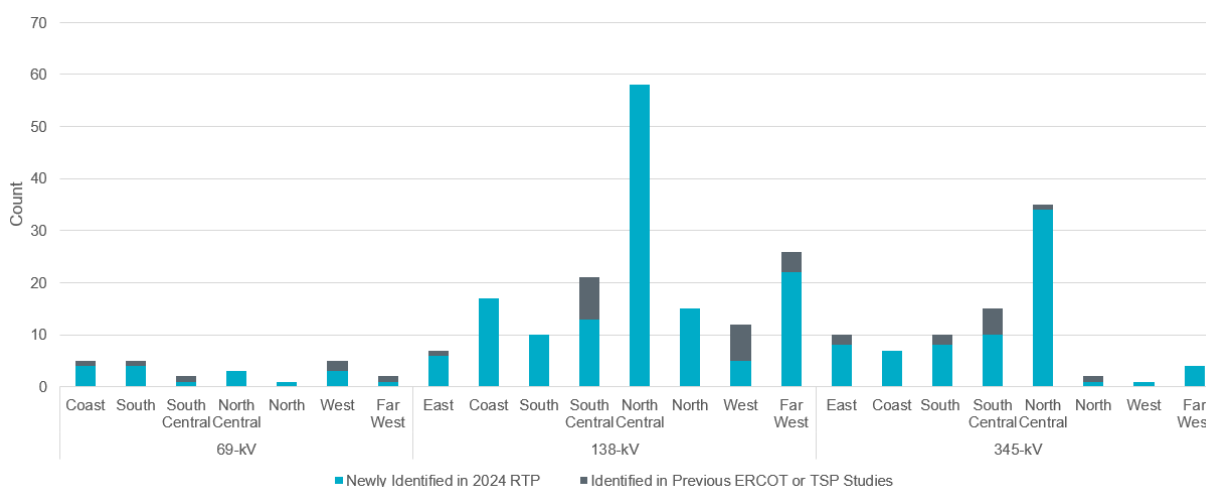


Figure 12: Projects Newly Identified in the 2024 RTP versus Projects Previously Identified

ERCOT, in collaboration with TPs, also identified five potential CMPs as placeholder mitigating actions, which will be reviewed in the operations planning horizon by ERCOT and TOs. The list and details of the CMPs identified in the 2024 RTP can be found in Appendix G.

3.1.1 West Texas Study Findings

As described in Section 2.1, the Permian Basin load forecast from the S&P Global Commodity Insights (formerly IHS Markit) study was adopted in the 2024 RTP, similar to the 2021, 2022, and 2023 RTPs. This projected oil and gas load, along with large loads seeking interconnection, increased the Far West load to 18 GW by 2029 and above 20 GW by 2030 under summer peak conditions. In comparison, the 2029 summer peak load forecast for the Far West Weather Zone was under 15 GW in the 2023 RTP. In July 2024, ERCOT developed the Permian Basin Reliability Plan as directed by the PUCT to ensure the reliable serving of both oil and gas loads and large loads. The 2024 RTP incorporated the Permian Basin Reliability Plan projects needed by 2030 and addressed any additional reliability needs outside of the Permian Basin Reliability Plan study scope. All five 345-kV import paths

identified by the Permian Basin Reliability Plan were included in the 2030 summer peak case to facilitate the comparison between the 345-kV plan and the 765-kV plan.

3.1.2 Central Texas Study Findings

It was observed in the 2023 RTP that conventional Generation Resource retirements in and around the San Antonio area contributed to stress on the existing 345-kV Central Texas corridor as new renewable Generation Resources in the South Weather Zone made their way up to support load growth in the South Central Weather Zone. In the 2024 RTP, generation in both the South and South Central Weather Zones increased compared to the 2023 RTP cycle.

Despite generation increases in the South and South Central Weather Zones, the increase in large load interconnection requests outpaced the generation growth. The new large load pockets in the South and South Central Weather Zones have changed the flow patterns from what had been observed in the 2023 RTP. There was an addition of approximately 15.3 GW of load in the 2030 summer peak case primarily in the following counties: Victoria, Nueces, and Cameron. The approximate large load breakdown by county is illustrated in Figure 13.

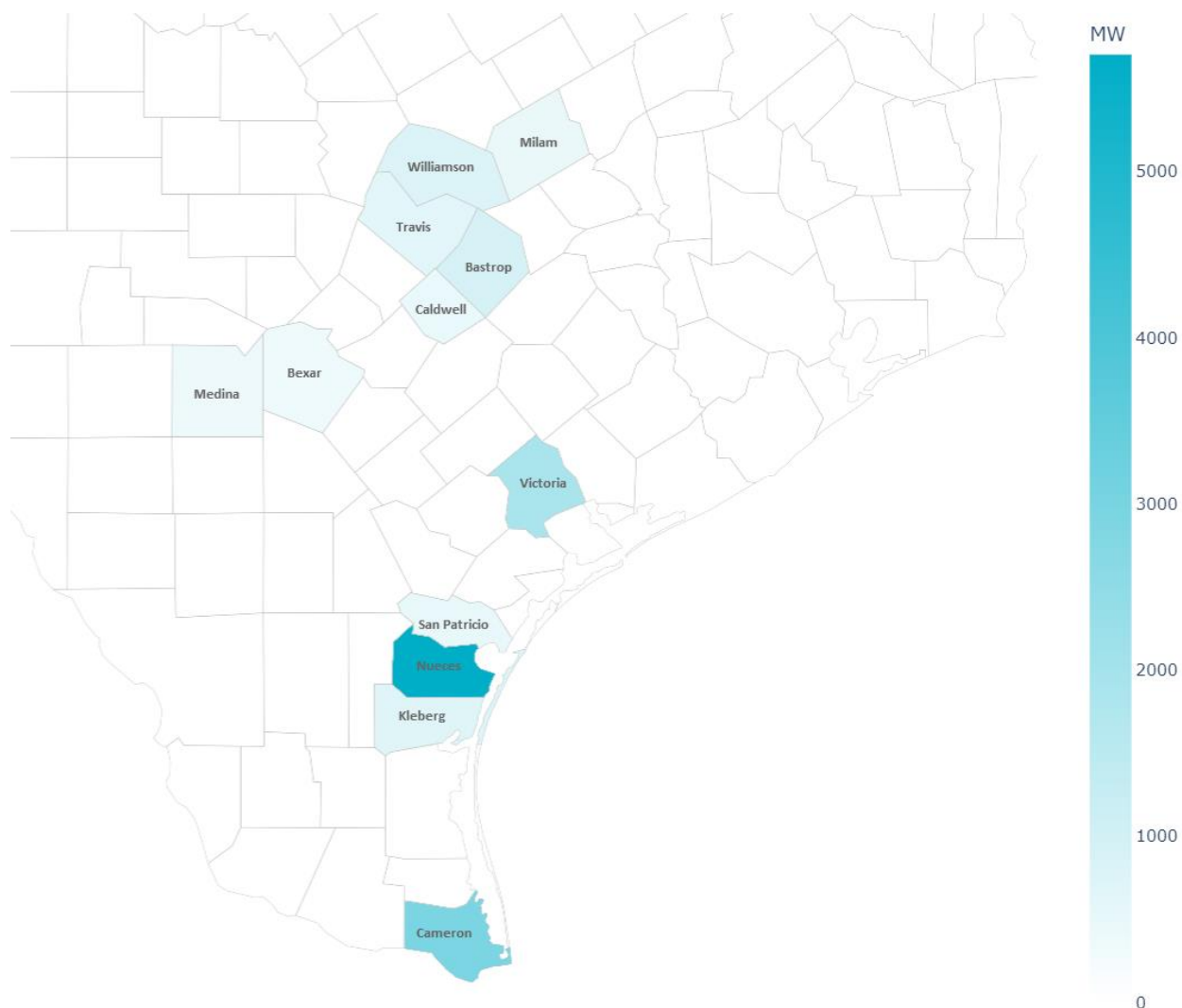


Figure 13: South and South Central Load Pocket Breakdown by County

Renewable resources in the South Weather Zone are no longer exporting power to the San Antonio and Austin areas as they are now serving the large loads in Victoria, Nueces, and Cameron counties. However, Central Texas still faces the challenge of needing additional import capability. To meet the higher projected load demand in Central Texas, new flow patterns have emerged, with imports to Central Texas coming from north of Austin and the Coast Weather Zone. Instead of requiring an additional import path from South to Central Texas, as proposed in the 2023 RTP, it was found in the 2024 RTP that reinforcing the Coast to South Central import paths was necessary to alleviate stress on the existing corridors. The three import paths coming from the Coast Weather Zone to Central Texas are illustrated in Figure 14. In addition, there was a need for additional import capacity along the pathway from north of Austin to Central Texas as illustrated by Figure 15. A need to have more 345/138-kV transformer capacity in Central Texas to serve the load at the 138-kV voltage level was also observed in the 2024 RTP.

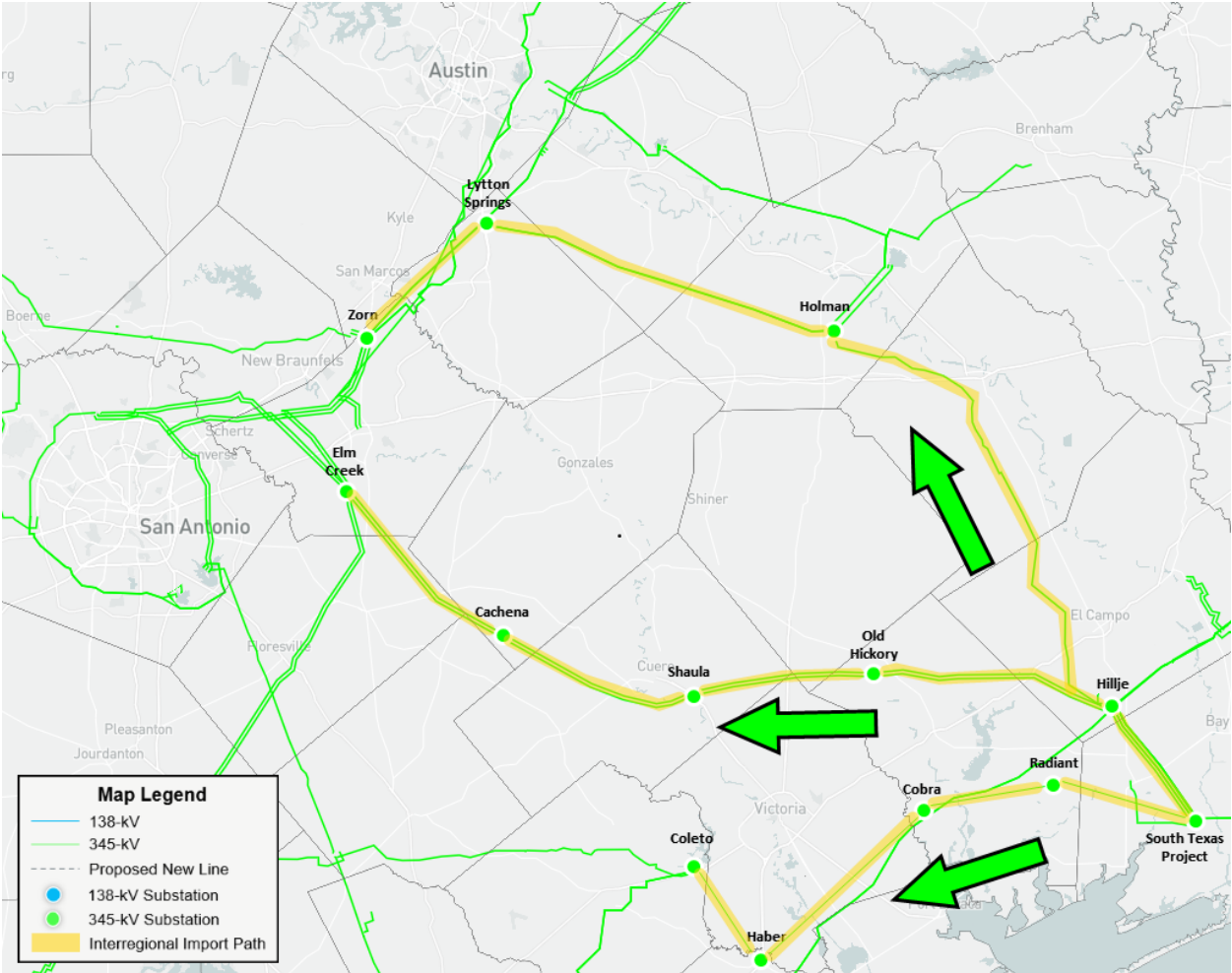


Figure 14: Coast to South Central Import Path

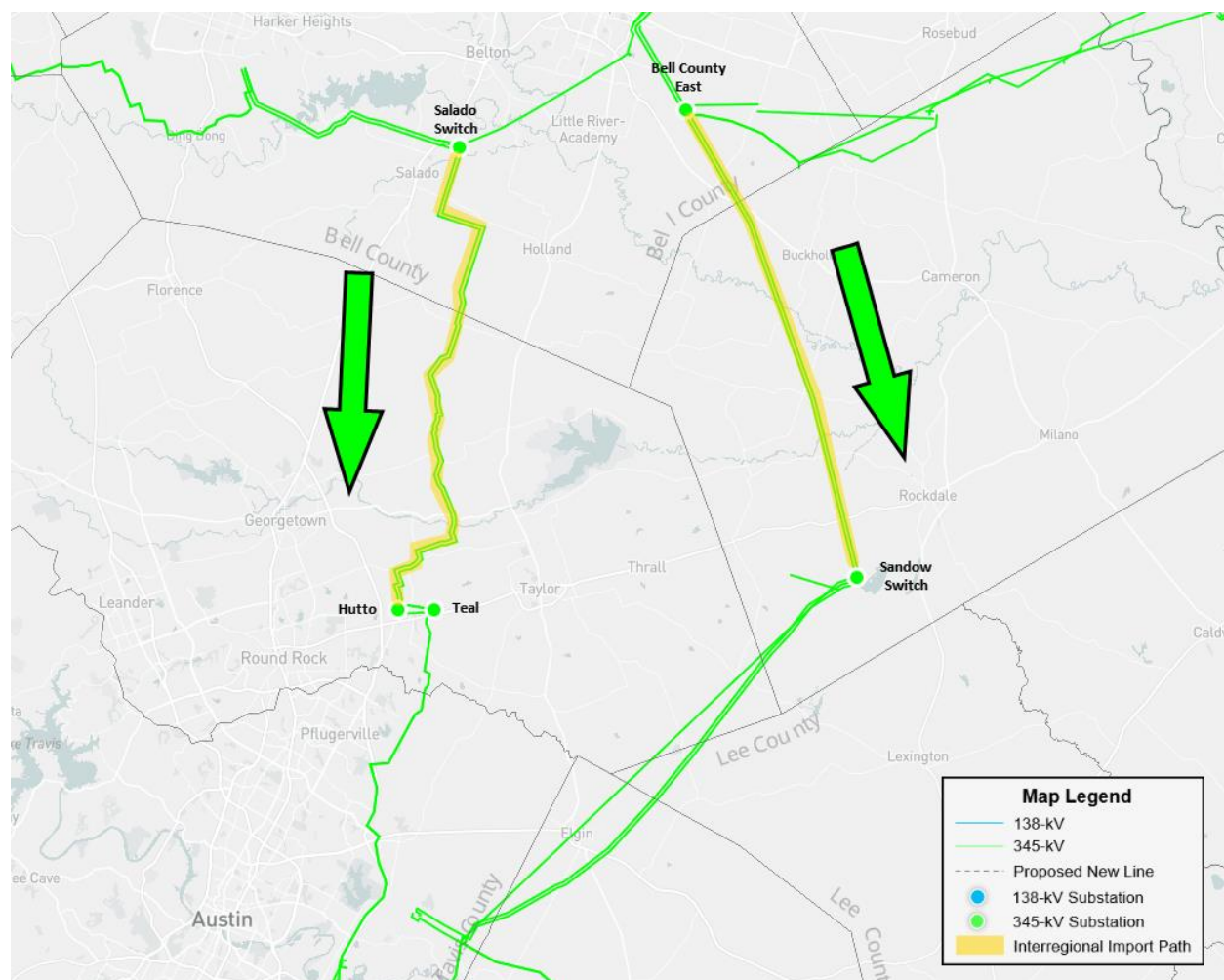


Figure 15: Central Texas Import Path from North of Austin

The detailed description of placeholder projects along both import paths can be found in Appendix E. Expected future generation and load will likely impact these needs, and ERCOT and TSPs will continue to evaluate the need and additional project options.

3.1.3 South Dallas Study Findings

As demonstrated in Figure 5, Location of Large Loads in 2024 RTP, in Section 2.1 of this report, Dallas County has significantly more large loads seeking interconnection than any other county in Texas. New load modeling assumptions dictated by HB5066 resulted in the addition of approximately 6.9 GW of load demand concentrated within the southern region of Dallas County for the 2029 summer peak case. This number increased to approximately 8.0 GW in the same region for the 2030 summer peak case.

The concentration of a large amount of load in a relatively small area brought substantial challenges to the transmission network. In addition to the significant need for local voltage support, it also placed tremendous stress on the existing import paths to the area. Additional import paths and reinforcement of the existing import paths were needed to ensure sufficient generation flow into the area to support

the loads. In total, approximately 580 circuit miles of new 345-kV import paths south of the Dallas-Fort Worth metroplex, along with hundreds of miles of reinforcement to the existing 345-kV import pathways and 5,000 MVAR of reactive power support, were needed to ensure the system remains secure. The initial study results from the area TSPs indicated that part of the reactive power support needs had to come from dynamic devices. ERCOT and the area TSPs will continue to review the needs in this area to reliably interconnect the requested large loads. Dynamic analysis is needed to identify the portion of the 5,000 MVAR reactive power support that must come from dynamic reactive power devices. The proposed new import pathways to South Dallas are illustrated in Figure 16. ERCOT and the area TSPs will continue to work on the needed transmission projects in this area in the 2025 RTP.

Portions of the identified transmission needs will be phased into earlier years given the staggered interconnection of large loads in the area.

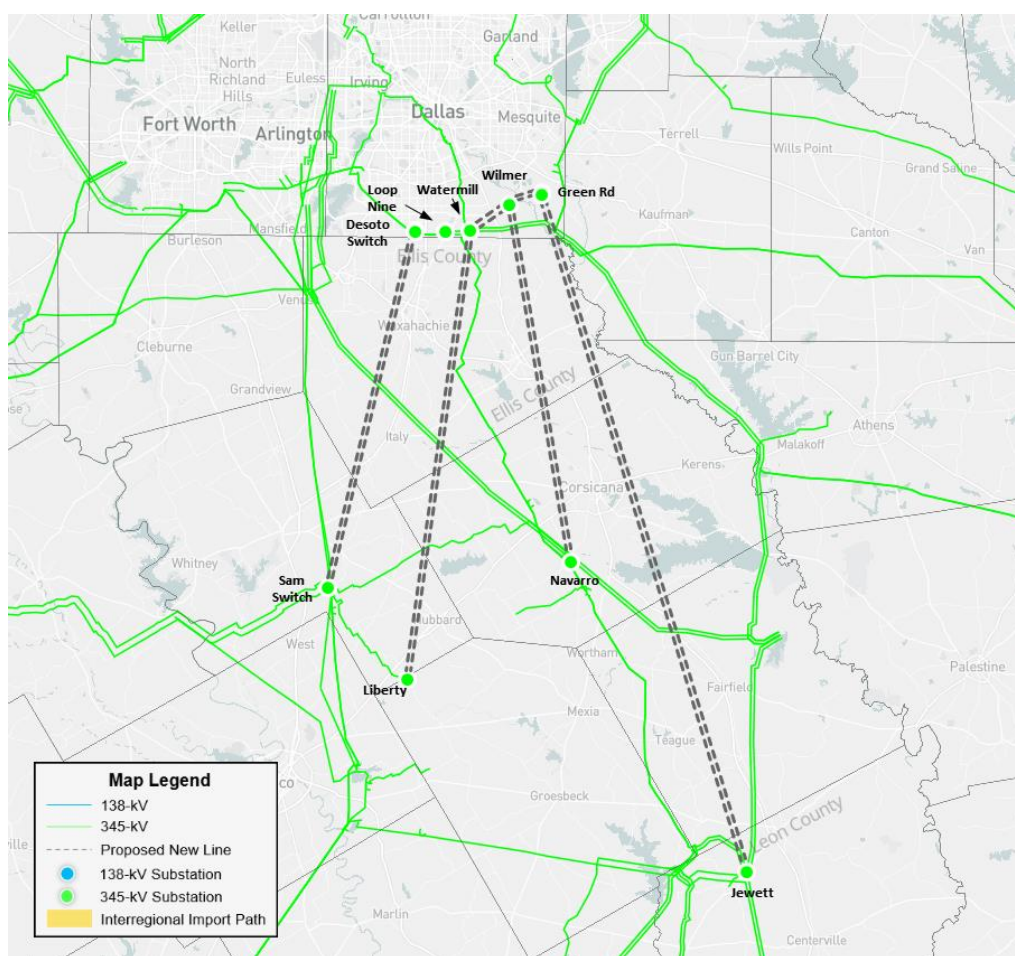


Figure 16: Placeholder New Pathways to South Dallas

The detailed description of the placeholder projects to address the identified reliability needs can be found in Appendix E. Expected future generation and load will likely impact these needs, and ERCOT and TSPs will continue to evaluate the need and additional project options. Oncor has submitted the Tier 1 Venus Switch to Sam Switch 345-kV Line project (24RPG017) and the Tier 1 Wilmer 345/138-kV Switch project (24RPG022) in 2024 for RPG review as the first step to address reliability issues

introduced by the interconnection of large loads in the area. Both projects are currently under ERCOT independent review. More RPG submissions are expected to follow to ensure the reliable interconnection of load in the area.

3.1.4 Generation Changes in the Coast Weather Zone

Historically, the Coast Weather Zone has primarily imported power to supply load in the region, especially in the Houston metroplex area. Stress on the import pathways both north and south of Houston has been observed in past RTPs. Unlike previous RTPs, nearly 30% of the planned Generation Resources that met Planning Guide Section 6.9 requirements and were added in the 2024 RTP are located in the Coast Weather Zone, mostly concentrated south of Houston and some just north of Houston. The accelerated generation development in the Coast Weather Zone, coupled with significant load growth in Central Texas, South Dallas, and Corpus Christi, transformed the Coast Weather Zone into an exporting weather zone. This shift demonstrates how rapidly evolving grid conditions can completely change the system flow patterns. TSPs serving Coast Weather Zone load have discussed with ERCOT potential large load growth in the Houston area, which may be reflected in future RTP studies, possibly further changing the flow patterns observed in the 2024 RTP.

A large portion of this new generation is expected to flow from Coast to neighboring weather zones to serve large load increases. New solar and battery generation located south of Houston flows further south to the South Texas Project and Hillje substations, then to the South and South Central Weather Zones. New generation north of Houston flows west to Central Texas and north towards Dallas County to serve the large load increases in those areas. This substantial generation export causes many 345-kV lines in the Matagorda and Wharton counties to consistently overload. New 345-kV export pathways, as well as upgrades to existing 345-kV pathways, were both needed to resolve these export challenges.

The detailed description of the placeholder projects can be found in Appendix E. Expected future generation and load will likely impact these needs, and ERCOT and TSPs will continue to evaluate the need and additional project options.

3.1.5 Other Findings

In addition to the reliability analysis summarized in previous sections, a multiple element outage analysis was conducted for contingencies where non-consequential load loss is allowed under NERC Reliability Standard TPL-001-5.1, Table 1. This consisted of:

- corrective action analysis, which identified mitigation measures (such as transformer tap setting changes, switching actions, generator re-dispatch, and load shed) to resolve any overloads and over/under-voltage issues resulting from such contingencies; and
- cascading analysis, which identified any contingencies that could result in potential cascade events.

Some planning events and extreme events were screened for detailed analysis, and further investigation performed by ERCOT indicated that none of those events resulted in cascading

conditions. ERCOT also studied the loss of multiple generating stations due to the disruption of gas pipelines. The results of the multiple element outage analysis are documented in Appendix H. This appendix includes the list of critical contingencies identified as a result of this analysis and CAPs or recommendations necessary to mitigate the impact of these contingencies.

ERCOT also performed an analysis of known outages of generation and transmission facilities planned in the near-term planning horizon per Requirement 2.1.4 of NERC Reliability Standard TPL-001-5.1. ERCOT issued Market Notices¹¹ to collect known outages information from both TOs and GOs. TOs were required to provide technical rationales for their known outages selection for each study case included in the 2024 RTP. ERCOT developed the technical rationale for the selection of GO-submitted outages to be included in the corresponding study cases. The known outages from the TOs and the GO outages selected based on the ERCOT-developed technical rationale were then incorporated into the base cases to study their impact on system performance under P0 and P1 contingencies. The study results concluded that no additional violations were caused by the known outages.

For the minimum deliverability analysis, ERCOT used the 2030 summer peak case as the start case for the analysis. The analysis found that additional transmission upgrades were needed to ensure the deliverability of the defined Generation Resources. Detailed information can be found in Appendix N.

ERCOT created a 2030 fall peak case to study the maintenance outage reliability criteria and identified reliability issues under the studied system conditions. CAPs were proposed to address those issues and ensure sufficient outage windows for equipment owners to perform necessary maintenance. Detailed information on the reliability issues and CAPs can be found in Appendix O.

In addition to the above analyses, per ERCOT Planning Guide Section 3.1.1.2(3), the 2024 RTP analysis included a list of transmission facilities that were loaded above 95% of their applicable ratings under normal and contingency events (loss of single generating unit, transmission circuit, transformer, or common tower outage). This list is attached to the report as Appendix I.

3.2. 765-kV Plan Findings

The unprecedented load growth has introduced significant challenges to the ERCOT grid, marking a new era of planning. It is crucial to review and update current planning processes to adapt to these changes. In 2024, ERCOT has been exploring the potential introduction of a 765-kV voltage class to the ERCOT region. This higher voltage class is expected to provide a long-term, holistic solution to accommodate the rapid evolution of the ERCOT system. The benefits of higher voltage classes may include the following:

- Increased transfer capability to load centers;
- Flexibility on Generation Resource siting;
- Outage coordination capacity;

¹¹ https://www.ercot.com/services/comm/mkt_notices/W-B041824-01;
https://www.ercot.com/services/comm/mkt_notices/W-A041824-01.

- Reduced impact to Texas consumers due to less Right of Way requirements;
- Lower line losses;
- Possible retirement of series compensation devices; and
- Potential exit strategy for some current Generic Transmission Constraints.

In the 2024 RTP, ERCOT began by developing a holistic 765-kV plan and then identified the critical components (core plan) needed by 2030. After establishing the core plan, ERCOT developed a 765-kV plan in parallel with the regular 345-kV plan to address the reliability issues identified in the analysis. The RTP 765-kV plan was created using the 2030 summer peak case, which also served as the start case for the 345-kV plan. The start case for the 765-kV plan incorporated all local projects needed by 2030 in the Permian Basin Reliability Plan, along with the three Permian Basin 765-kV import paths. Analyses were performed under the N-1 contingency conditions, prior outage of a generator or 345/138-kV transformer followed by N-1 contingency conditions, and maintenance outages at the 300-kV and above voltage level. Figures 17 and 18 below illustrate the holistic 765-kV plan, broken down into two phases for future growth, and the core plan, respectively. ERCOT intends to file the 345-kV vs 765-kV comparison plan based on the 2024 RTP together with the dynamic analysis and congestion analysis with PUCT in January 2025.

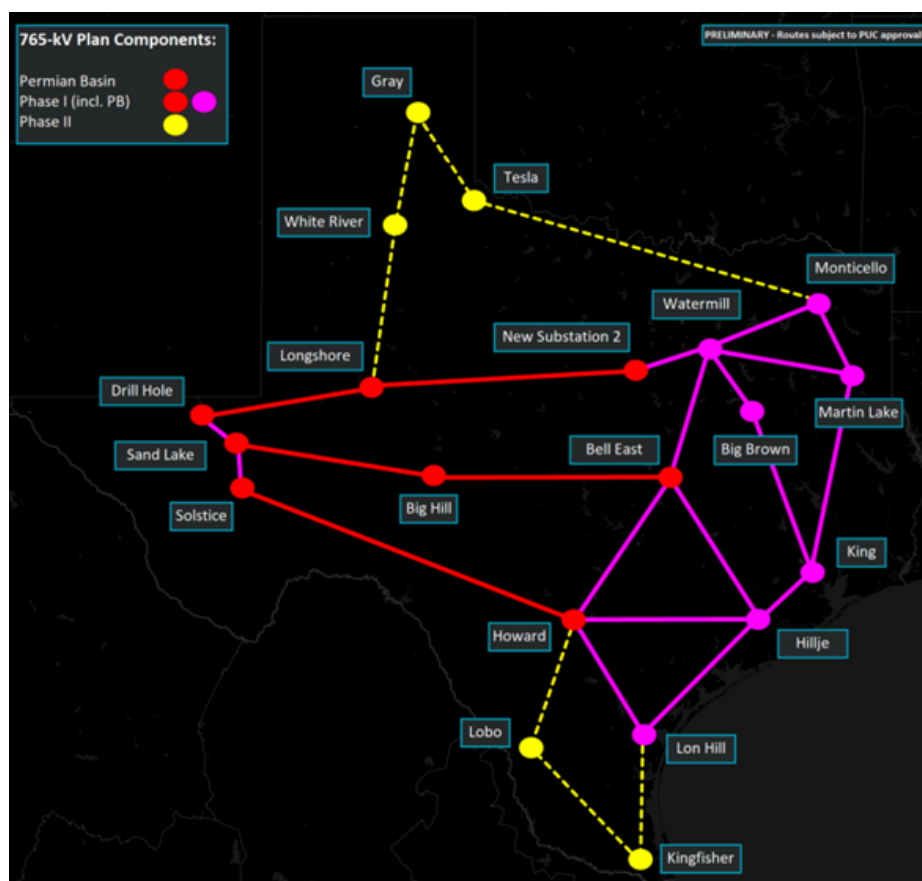


Figure 17: The Holistic 765-kV Plan in Two Phases

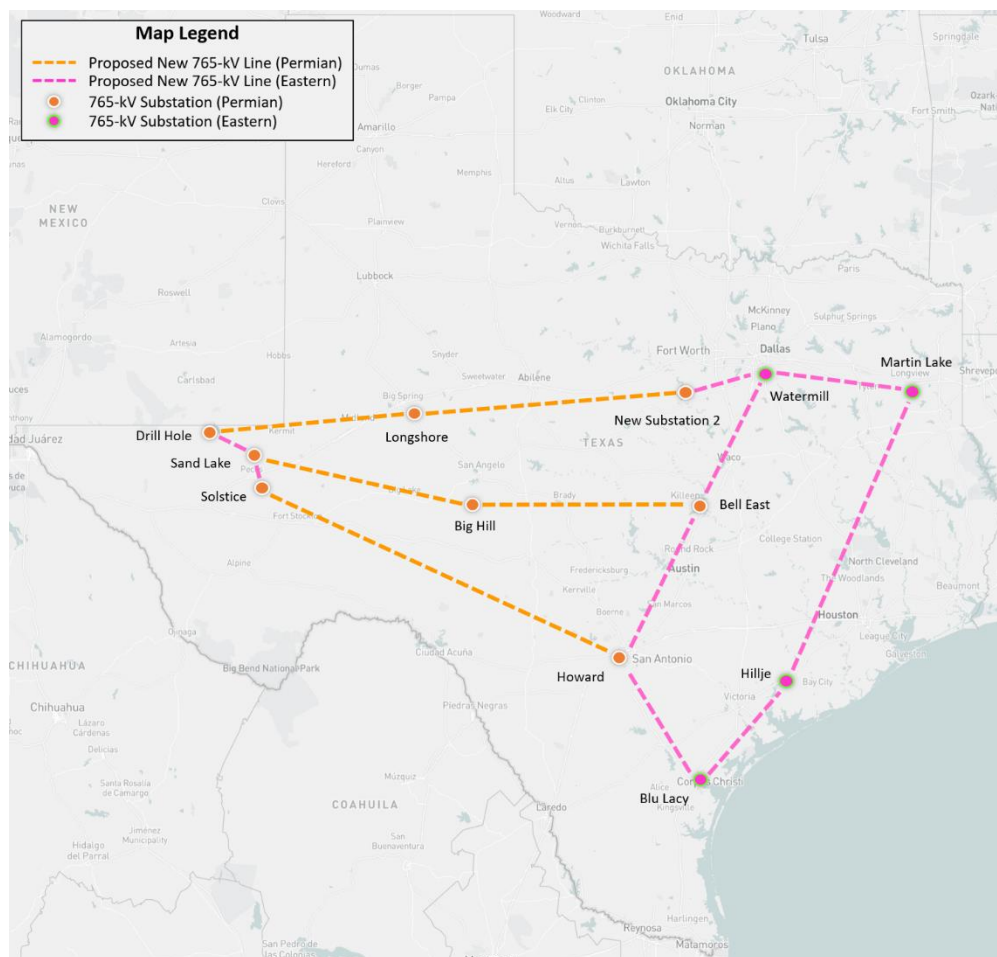


Figure 18: The 765-kV Core Plan needed in 2030

3.3. Sensitivity Analysis

The unprecedented load amount incorporated in the 2024 RTP necessitated going beyond the current planning practices to add generation to meet the demand. Some of this generation is not in the current interconnection queue. Stakeholders have raised two key questions:

1. How will system needs change if not all large loads materialize?
2. What are the impacts of the added generation, which are not in the current interconnection queue, on system needs?

The sensitivity analysis was designed to answer these questions. In the sensitivity analysis, all large loads substantiated by TSP officer letters were reduced to 50% and all generators not currently in the interconnection queue were removed from the cases. The sensitivity analysis was performed on the 2026 and 2029 summer peak cases, and the 2027 minimum load case. Despite reducing the load substantiated by TSP officer letters from over 31 GW to less than 16 GW, the key reliability needs identified in the base case reliability analysis remained similar. However, the addition of generation not in the current interconnection queue may have masked certain reliability needs. After removing two large generators in the Permian Basin area, it was found that additional import capacity to the area

was needed, and the stage 5 project identified by the ERCOT Delaware Basin Load Integration Study was required to provide these import needs.

ERCOT is currently working on a Planning Guide Revision Request to address this particular change in the planning process, brought about by the new era of planning. Stakeholders have also begun discussions on additional policy changes governing the inclusion of loads in the planning models.

The detailed assumptions and study results are available in Appendices B and J, respectively.

3.4. Short Circuit Analysis

As indicated in Section 2.2, ERCOT conducted short-circuit analysis for Resource Entity-owned facilities for 2027 summer peak conditions based on the system protection future year base case and shared the results with GOs. GOs reviewed the fault duty information to identify buses with over-dutied breakers and CAPs.

Table 1 provides a summary of the results of the short-circuit analysis. The study cases and details of the results can be found in Appendix K.

Table 1: Summary of Short-circuit Analysis

Magnitude of Fault Current	Number of buses (3-phase fault)	Number of buses (single-line-to-ground fault)
Below 40 kA	574	587
40 kA ~ 60 kA	82	67
More than 60 kA	4	6

3.5. Long Lead Time Equipment Analysis

In response to ERCOT's request, TOs provided a list of long lead time equipment based on their spare equipment strategies. All TO-provided BES long lead time equipment outages were studied to determine the impact of unavailability of such equipment for an extended period of time. This analysis was conducted for the 345-kV plan for 2026 and 2029 summer peak conditions, along with 2027 off-peak conditions. Overall, 37 unique 345-kV transformers, 21 unique 345-kV reactive devices, 1 unique reactive device at other voltage levels, 4 345-kV synchronous condensers and 2 of their transformers, 2 unique 138-kV STATCOMs, 1 unique 345-kV breaker, 3 unique 345-kV SVCs, and 4 unique 138-kV SVCs were identified as long lead time equipment. NERC category P0, P1, and P2 planning events were studied. The results were shared with the respective TPs. The list of long lead time equipment and study results are provided in Appendix L.

4. Economic Analysis

The 2024 RTP economic analysis was performed using production cost simulation for years 2026 and 2029. In the analysis, both the PCS test and GRR test were utilized. Through this assessment, ERCOT identified transmission congestion and tested various transmission improvements to address this congestion in a cost-effective manner (as defined by ERCOT's economic planning criteria). Eight economic transmission improvement projects were evaluated in the 2024 RTP. In addition to the PCS test and the GRR test, the CCS test results were also provided for informational purposes only at this time. As noted in section 2.5, NPRR1247, which proposes to add the CCS test to the ERCOT Protocols, was recommended for approval by the ERCOT Board of Directors in December 2024 and is anticipated to be considered by PUCT for approval in January 2025. Break-even capital cost was provided for the projects if the test results showed positive benefits.

The input data and congestion tables from the 2024 RTP can be found in Appendices C and D. Table 2 provides a system summary of 2024 RTP economic analysis for years 2026 and 2029.

Table 2: System Summary of 2026 and 2029¹²

Description	Unit	2026	2029
Coincident Peak Load	MW	90,702	94,410
Peak Net Load ¹³	MW	72,285	78,477
Annual Served Demand	GWh	537,766	581,538
Annual Storage Charging	GWh	6,184	6,825
Annual Transmission Losses	GWh	12,880	14,523
Annual Generation	GWh	556,830	602,886
Load-Weighted Average LMP	\$/MWh	25.67	28.21

While there may be hours when all ERCOT demand could theoretically be served by wind and solar resources, thermal and stability constraints on the transmission system and unit commitment limitations caused the grid simulation software to curtail available wind and solar output. Figures 19 and 20 summarize monthly production and curtailment for wind and solar generation, respectively.

¹² All results are based on the 2013 historical weather conditions.

¹³ Hourly Net Load = Hourly Load Forecast – Hourly Wind Output – Hourly Solar Output.

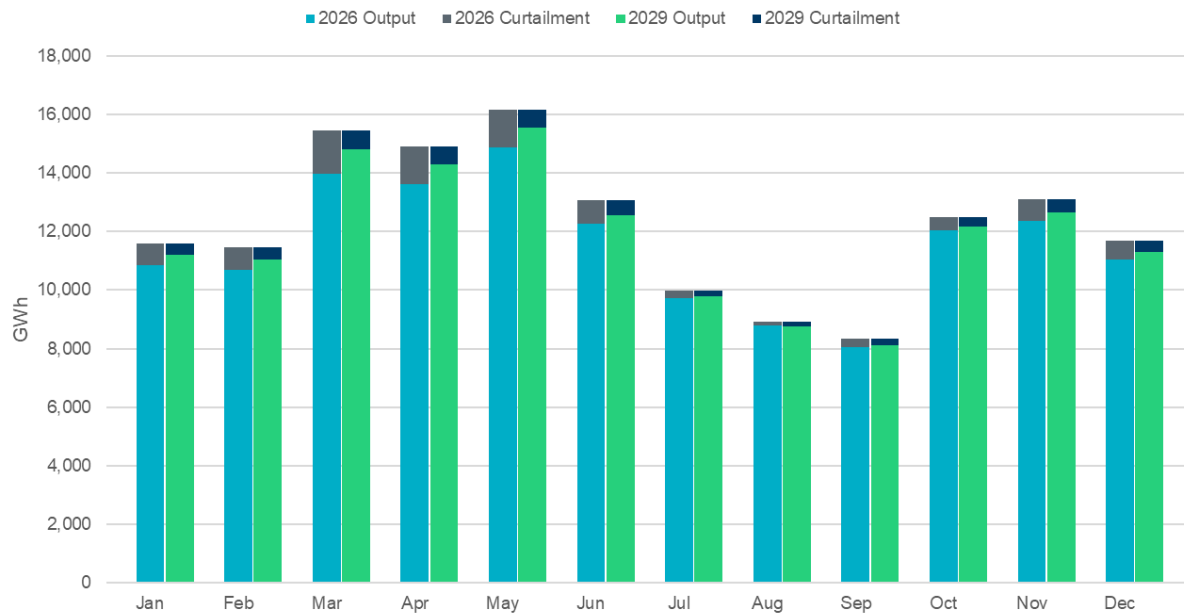


Figure 19: Wind Production and Curtailment

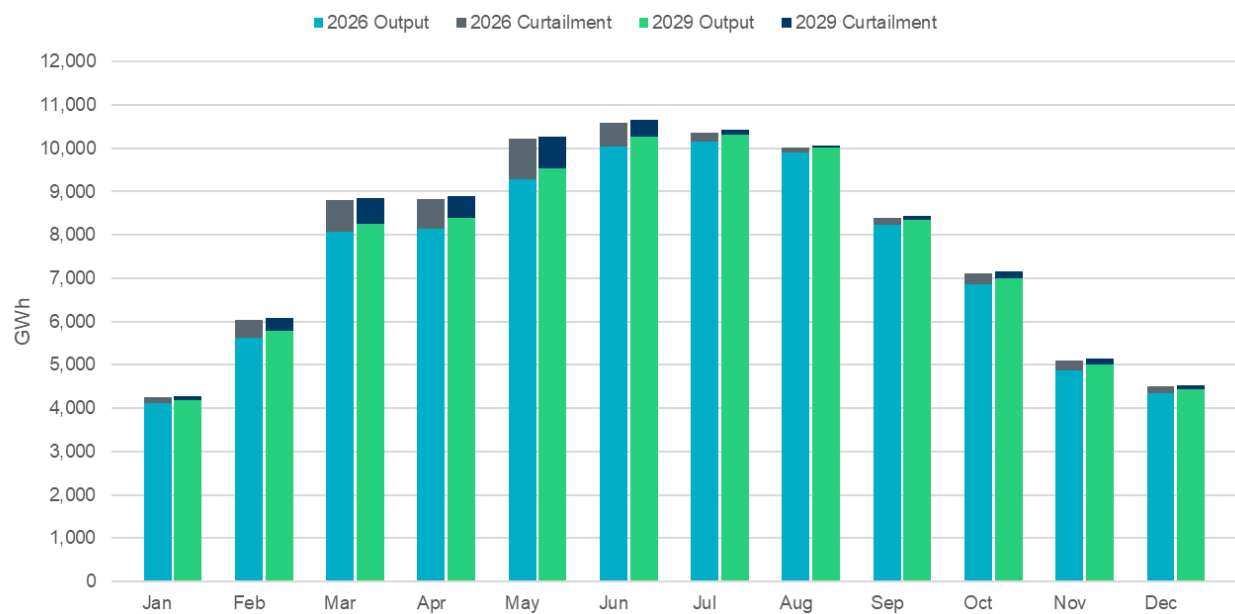


Figure 20: Solar Production and Curtailment

Figures 21 and 22 show the top constraints seen in 2026 and 2029, respectively. The size of each bubble represents the relative capacity of each congested element over the study period.

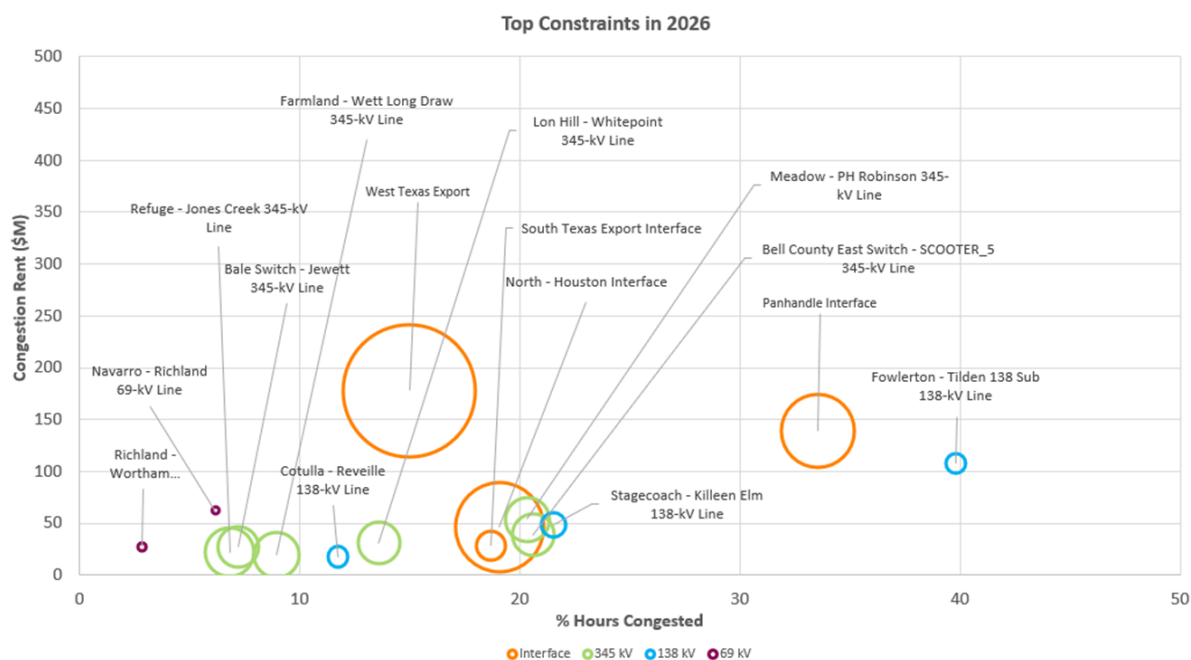


Figure 21: Top Constraints in 2026

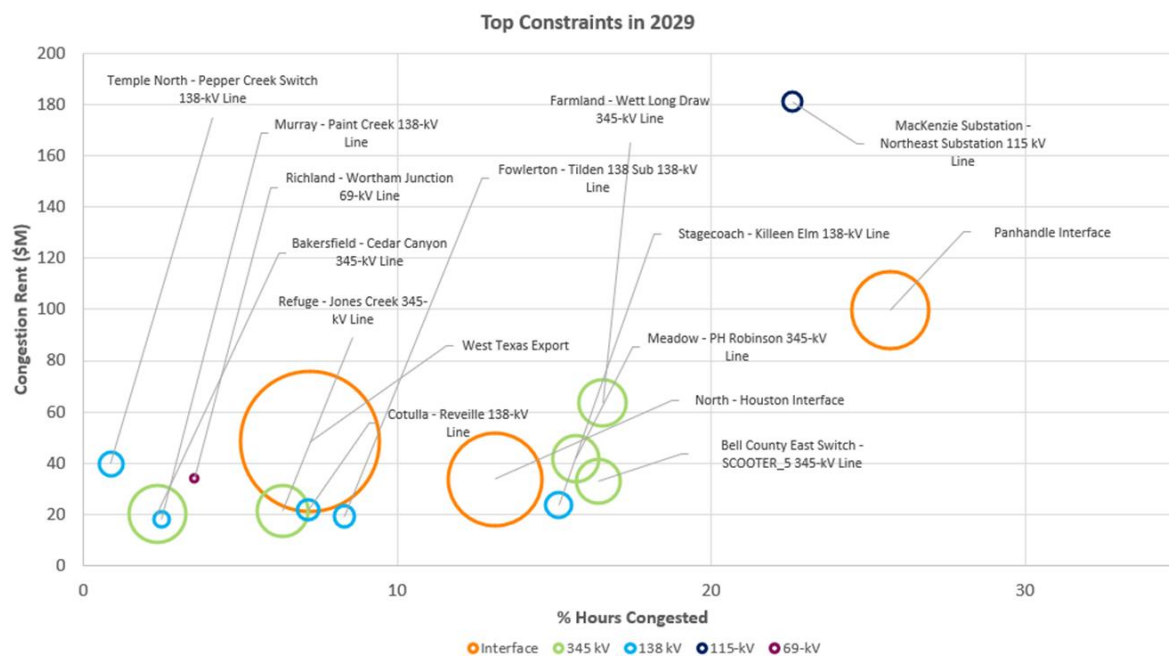


Figure 22: Top Constraints in 2029

Similar to the 2023 RTP economic analysis, the West Texas Export interface was the most congested element observed for both the 2026 and 2029 study years. The interface limit used in the 2024 RTP was 11,016 MW based on preliminary results from ERCOT's Long-Term West Texas Export Special

Study¹⁴. The interface was congested approximately 14.99% and 7.21% of hours in the 2026 and 2029 study years, respectively. Driven by the load growth in West and Far West Texas, the congestion cost for West Texas Export interface was less in 2029 (\$49 million) compared to 2026 (\$178 million).

Congestion rent of \$139 million and \$100 million was observed over the Panhandle interface in 2026 and 2029, respectively, due to lack of sufficient transmission capacity to relieve this constraint. To effectively alleviate congestion on the West Texas Export interface and/or Panhandle interface, the long-distance high voltage or direct current (DC) transmission lines are more favorable. However, these options are also expensive. A number of projects have been evaluated in 2024 RTP economic analysis and will be continuously analyzed in the future RTP and LTSA studies.

The North to Houston interface limit was modeled with hourly profiles based on historical data in the 2024 RTP economic analysis. Modest congestion was observed in both the 2026 and 2029 study years on the North to Houston interface, with the interface congested 19.05% and 13.09% of hours in the 2026 and 2029 study years, respectively. The congestion was driven by the increased renewable integration in North Central, East, and North Weather Zones. These results are consistent with real-time congestion on the North to Houston interface throughout 2024.

High congestion rent was observed in the Coast Weather Zone in both the 2026 and 2029 study years. This is attributed to a combination of the addition of solar generation southwest of Houston, the increased new renewable generation in the South Weather Zone, and the load growth in the Houston area. The Meadow to PH Robinson 345-kV line was congested 20.94% and 16.26% of hours in the 2026 and 2029 study years, respectively.

A notable change in 2024 RTP is the heavy congestion of the 115-kV network in the Lubbock area. Congestion rent of \$15 million and \$181 million was observed over the MacKenzie Substation to Northeast Substation 115-kV line in 2026 and 2029, respectively.

Due to renewable generation increase in the Lower Rio Grande Valley (LRGV) area, the North Edinburg to Lobo interface and Nelson Sharpe to Rio Hondo interface experienced high congestion in both real-time operations and the 2024 RTP economic study. It was observed that the North Edinburg to Lobo interface was congested 22.31% of hours in the 2026 study year. The Nelson Sharpe to Rio Hondo interface was congested 22.48% of hours in 2026. These observations are consistent with the findings identified by the 2023 RTP. Congestion in the area was driven primarily by the contingency involving the common-tower loss of the North Edinburg to Bonilla 345-kV line and the 138-kV line from Bonilla to South Santa Rosa. The primary 345-kV path was removed as part of the contingency, and the result was heavy congestion along the parallel 138-kV path to the west.

The ERCOT Board of Directors endorsed the LRGV System Enhancement Project in 2021. The PUCT also ordered the construction of a new second circuit on the double-circuit capable 345-kV transmission line that runs from San Miguel to Palmito and new transmission facilities to close the loop from Palmito to North Edinburg. These projects plan to be in operation before the summer of 2027.

¹⁴ <https://www.ercot.com/files/docs/2022/01/14/Long-Term-West-Texas-Export-Study-Report.pdf>.

Those improvements help to relieve the congestion on the North Edinburg to Lobo interface and Nelson Sharpe to Rio Hondo interface in 2029.

The South Texas Export (STE_PASP) and Import interface (STI_PASP) was congested 18.68% and 0.18% of hours in the 2026 and 2029 study years, respectively. The implementation of the following transmission projects is expected to help exit the South Texas stability constraints¹⁵:

- San Antonio South Reliability Project (endorsed, in-service date: 2027)
- LRGV System Enhancement Project (endorsed, in-service date: 2027)
- San Antonio South Reliability II Project (endorsed, in-service date: 2029)
- Tango-Goddard-Kateon-Lon Hill 345-kV Line second circuit (part of the South to Central Texas project from the 2023 RTP).

Finally, as required by paragraph (3) of ERCOT Protocols Section 3.10.8.4, ERCOT identified additional transmission elements that have a high probability of providing significant added economic efficiency to the ERCOT market using dynamic ratings. Dynamic ratings for the identified elements (listed in Appendix M) have been requested from the associated TOs.

4.1. Study Assumptions and Methodology

Pursuant to the amended 16 TAC § 25.101(b)(3)(A)(i), an economic cost-benefit study for economic projects must be performed under a CCS test and a PCS test. Until the CCS test becomes effective, § 25.101(b)(3)(A)(i)(I)(-b-) allows ERCOT to use the GRR test, which was used for evaluation of economically driven projects in the ERCOT Region during the 2006 to 2012 timeframe as the congestion cost savings test. To pass the PCS test, the levelized ERCOT-wide annual PCS attributable to the proposed project should be equal to or greater than the first-year annual revenue requirement (12.9%) of the proposed project. To satisfy the GRR test requirement, the levelized ERCOT-wide annual GRR attributable to the proposed project should be equal to or greater than the average of the annual revenue requirement for the first three years (12.6%) of the proposed project. These revenue requirements are reviewed annually and may vary from year to year. ERCOT may recommend, and the PUCT may approve, a transmission upgrade in the ERCOT Region that passes either a CCS test (in the interim, a GRR test) or a PCS test. The total production cost is the sum of the fixed operation and maintenance (O&M), startup, variable O&M, fuel, and emission cost of generators. The total generator revenue is equal to the sum of the energy production of the generator multiplied by its nodal price. In addition to the results of the PCS test and the GRR test, results for the CCS test were also provided for informational purposes only at this time.

4.2. Top Constraints for 2026 and 2029 Study Years

The economic analysis continued to demonstrate significant congestion for both the 2026 and 2029 study years. Based on the review of the initial congestion pattern and stakeholder feedback, ERCOT

¹⁵ https://www.ercot.com/files/docs/2024/07/25/05-gtc_update_stx-export-and-import-aug1-ros-3.pdf.

selected transmission projects to conduct both the PCS test and the GRR test, as guided by the amended 16 TAC § 25.101(b)(3)(A)(i) outlined in section 4.1.

Table 3 shows the projected top 10 constraints ranked by the congestion rent on the ERCOT System based on the economic analysis conducted for the study years 2026 and 2029. Figure 23 shows locations of the top 10 constraints in 2026 and 2029.

Table 3: Top Congested Constraints from 2026 and 2029 Study Years

Index	Constraint	Congestion Rent ¹⁶ (M\$)	
		2026	2029
1	MacKenzie Substation - Northeast Substation 115-kV Line	15	181
2	West Texas Export Interface	178	49
3	Panhandle Interface	139	100
4	Fowlerton - Tilden 138 Sub 138-kV Line	108	19
5	Farmland - Wett Long Draw 345-kV Line	19	64
6	Navarro - Richland 69-kV Line ¹⁷	62	-
7	Meadow - PH Robinson 345-kV Line	54	42
8	Stagecoach - Killeen Elm 138-kV Line	49	24
9	North - Houston Interface	46	34
10	Temple North - Pepper Creek Switch 138-kV Line	-	40

¹⁶ Congestion rent indicates areas of the system where economic transmission projects may be beneficial. It is not an indication of whether a project to reduce specific congestion would or would not meet the ERCOT economic planning criteria.

¹⁷ Navarro to Richland 69-kV Line Upgrade (2023-NC39) reliability project from the 2023 RTP (needed by summer 2028).

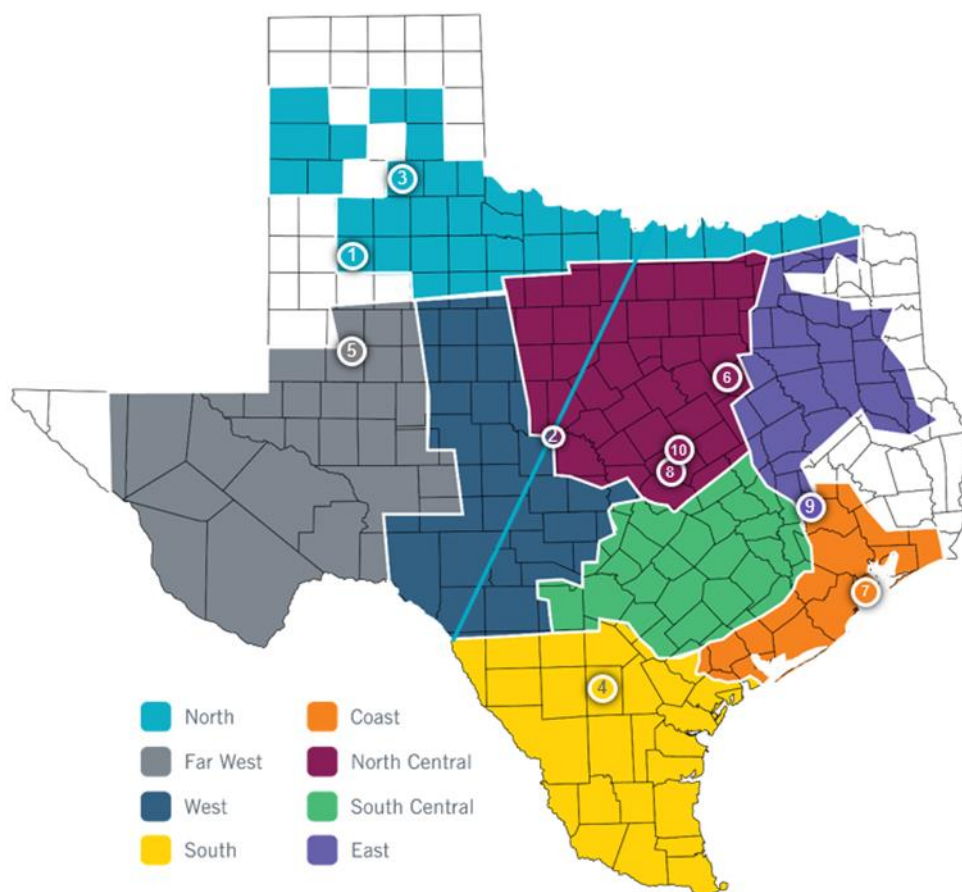


Figure 23: Locations of Top 10 Constraints in 2026 and 2029

4.3. Projects Selected for Evaluation

The congestion patterns observed for the 2026 and 2029 study years served as the basis for identifying potential economic projects for further evaluation.

In addition to the projected congestion outlined above, ERCOT also reviewed historical congestion observed in those areas and took experiences with these constraints in past economic models into consideration to select several additional projects for further evaluation.

Table 4 shows the list of all transmission projects evaluated in this economic analysis. Figure 24 shows the location of each project. Detailed descriptions of these projects are included in Appendix P.

Table 4: Projects Selected for Economic Evaluation

Index	Description
Project 1	Richland – Wortham 69-kV Line Upgrade
Project 2	Farmland Area Improvement (Upgrade Farmland – Long Draw and Farmland – Fiddlewood Switch 345-kV lines)
Project 3	Lubbock Area Improvement (Upgrade 115-kV transmission lines: Yellow House – Northwest – McDonald – Mackenzie – Northeast – Dunbar – Holly)

Project 4	Coast Area Improvement Option 1 (New 345-kV line: STP – Bailey and Bailey – PH Robinson)
Project 5	Coast Area Improvement Option 2 (Project 4 Plus New 345-kV line Bailey – Zenith, Upgrade Zenith – TH Wharton DCKT and PH Robinson – Meadow 345-kV lines)
Project 6	Murray – Paint Creek 138-kV Line Upgrade
Project 7	Lon Hill – White Point 345-kV Line Upgrade
Project 8	Bell County East Switch – Scooter 345-kV Line Upgrade

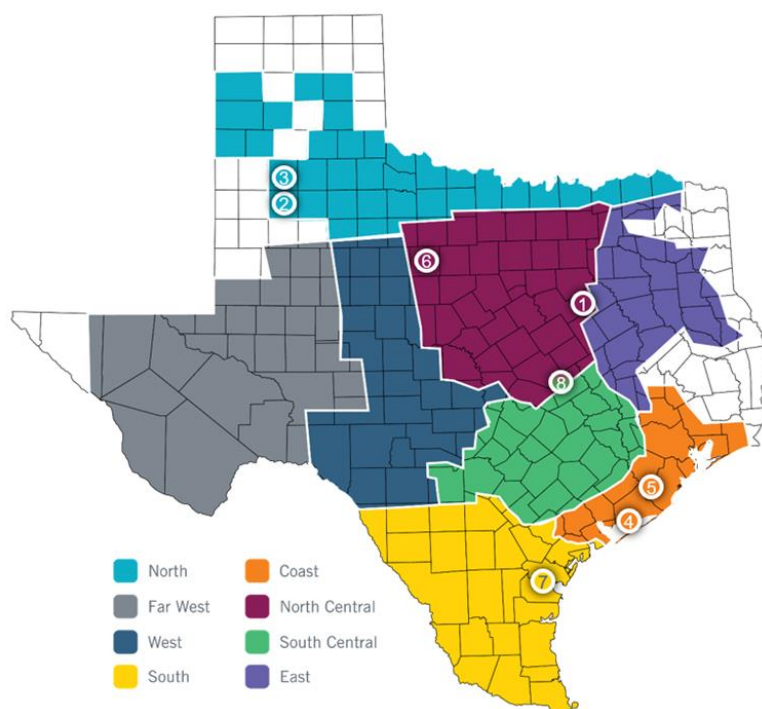


Figure 24: Proposed Project Locations

4.4. Project Evaluation Results

Using the updated 2024 RTP economic cases (both 2026 and 2029 study year cases), ERCOT performed an economic benefit-cost analysis for eight transmission enhancements chosen based on a review of the initial congestion patterns and historical data. For all transmission projects, both the PCS test and the GRR test were performed and a break-even capital cost, if applicable, was provided.

ERCOT evaluated, for informational purposes, the benefit/cost ratio for both the PCS and GRR tests using a generic cost estimate. Cost estimates from the TSPs will be used in the RPG review process. A transmission system upgrade is economically viable if 1) it meets the PCS test or 2) it meets the GRR test requirement.

As previously noted, in addition to the results of the PCS test and the GRR test, the CCS test results are also provided for informational purposes only at this time.

Details of the economic analysis can be found in Appendix P.

5. Appendices

Index	Description	Document	Access
A	RTP Scope and Process Document	Appendix_A_2024_RTP_Scope_and_Process_Final.pdf <file included in the public version>	Public
B	Input assumptions for the 2024 RTP reliability analysis	Appendix_B_2024_RTP_Reliability_Input_Assumptions.xlsx <file included in the public version>	Public
C	Input assumptions for the 2024 RTP economic analysis	Appendix_C_2024_RTP_Economic_Input_Assumptions.xlsx <file included in the public version>	Public
D	Economic analysis start case inputs and annual constraints	Appendix_D_2024_RTP_Economics_Start_Case_Inputs_Annual_Constraints.zip <file available in MIS Secure Area>	MIS Secure
E	Reliability Driven Projects – 345-kV	Appendix_E_2024_RTP_Reliability_Projects_345-kV_Plan_Public.xlsx <file included in the public version>	Public
F	Project locations	Appendix_F_2024_RTP_Project_Locations.pdf <file included in the public version>	Public
G	Constraint Management Plans	Appendix_G_2024_RTP_Constraint_Management_Plans.xlsx <file available in MIS Secure Area>	MIS Secure
H	Multiple element contingency analysis	Appendix_H_2024_RTP_MultipleElementContingencyStudyReport.docx <file is ERCOT-Confidential>	N/A
I	Facilities loaded over 95%	Appendix_I_2024_RTP_95%_Exceedance_PG31123.xlsx <file available in MIS Secure Area>	MIS Secure
J	Sensitivity Analysis Results	Appendix_J_2024_RTP_Sensitivity_Projects.xlsx <file available in MIS Secure Area>	MIS Secure
K	Short circuit Analysis	Appendix_K_2024_RTP_ShortCircuitStudyCases_DetailedResults.docx <file available in MIS Secure Area>	MIS Secure
L	Long lead time equipment analysis	Appendix_L_2024_RTP_LongLeadTimeEquipment.docx <file is ERCOT-Confidential>	N/A
M	Transmission elements proposed to be dynamically rated	Appendix_M_2024_RTP_Dynamic_Rating_NP3_10_8_4.xlsx <file available in MIS Secure Area>	MIS Secure
N	Minimum Deliverability Analysis	Appendix_N_2024_RTP_Minimum_Deliverability_Projects.xlsx <file available in MIS Secure Area>	MIS Secure
O	Maintenance Outage Analysis	Appendix_O_2024_RTP_Maintenance_Outage_Projects.xlsx <file available in MIS Secure Area>	MIS Secure
P	Economic Analysis Results	Appendix_P_2024_RTP_Economic_Analysis.pdf <file included in the public version>	Public
Q	Reliability Driven Projects – 765-kV	Appendix_Q_2024_RTP_Reliability_and_MaintenanceOutage_Projects_765-kV_Plan.xlsx <file available in MIS Secure Area>	MIS Secure